

Audit and development log: current-job tenure model

Reported start dates, job changes, and duration contamination

04 September 2026

Purpose and status

This document is a reproducible audit and development log for improvements to the tenure-contamination models reported in Table 7. It is deliberately separate from the branch-wide summaries. Its purposes are to:

1. document the provenance and construction of the duration variables;
2. record descriptive evidence used to select measurement assumptions;
3. distinguish observed evidence from latent-model interpretation;
4. list every new estimation script and the checks applied to it; and
5. preserve point estimates, diagnostics, and model comparisons as the model is extended incrementally.

The start-date audit, job-reset extension, and discrete-month likelihood repair are implemented and estimated separately from Table 7. The continuous-density version fails full-model recovery and is retained only as a diagnostic. The monthly probability-mass version passes repeated recovery checks, converges from dispersed starts, has regular local inference, and remains stable under stricter matching panels. It is the current candidate model, but the remaining measurement-process extensions below should be assessed before it replaces the paper’s Table 7 specification.

1. Variable provenance and reconstruction

The upstream mapped QLFS file contains `yearstart` (year commenced working), `monthstart` (month commenced working), survey `year`, quarterly `wave`, and the derived tenure variable `timeago`. The matched panels retain `period` and `tenure`, but not the start-year and start-month components.

For survey period (p), define the nominal interview year and month by

$$Y(p) = 2008 + \left\lfloor \frac{p-1}{4} \right\rfloor, \quad M(p) = 3\{(p-1) \bmod 4 + 1\}.$$

Thus the nominal interview months are March, June, September, and December. For every upstream record with valid start-year and start-month fields,

$$\text{tenure months} = 12\{Y(p) - Y^{start}\} + M(p) - M^{start}.$$

Consequently, the reported start month can be recovered without an upstream merge as

$$R_{it} = 12Y(p_{it}) + M(p_{it}) - g_{it},$$

where (g_{it}) is tenure in months. This is a reported current-job start month, not necessarily the date of the worker’s most recent entry into employment. The exact interview day is unavailable.

Audit cautions

- A changed start date may reflect a new job, inconsistent reporting, or an incorrect record link.
- A date strictly after the previous nominal interview is strong evidence of a job change, but dates in the previous interview month are ambiguous because interview days are unavailable.
- The matching algorithms affect which records are linked. Comparisons across panels A, B, and C are therefore essential.
- January concentration can reflect genuine seasonality, rounding, or a reporting/coding convention. The available labels do not identify which.

2. Descriptive start-date audit

The audit uses the paper’s age range (18–55), all available consecutive waves, survey weights, and the three alternative matching panels. It classifies each consecutive employed pair as an unchanged start date, a plausible job reset, an ambiguous boundary-month reset, or a revision to a date that predates the previous interview.

Table 1: Start-date audit: survey-weighted percentages

Measure	Panel A	Panel B	Panel C
Same reported start month	64.6	65.5	69.5
Plausible job-to-job reset	1.7	1.7	1.5
Boundary-month reset	0.9	0.9	0.8
Revised pre-existing date	32.8	31.9	28.3
Whole-year shift among revisions	20.7	21.0	21.6
Revised date repeated next wave	31.8	31.8	35.2
Plausible reset repeated next wave	33.7	33.9	38.1
Reported start month is January	26.2	26.2	26.2

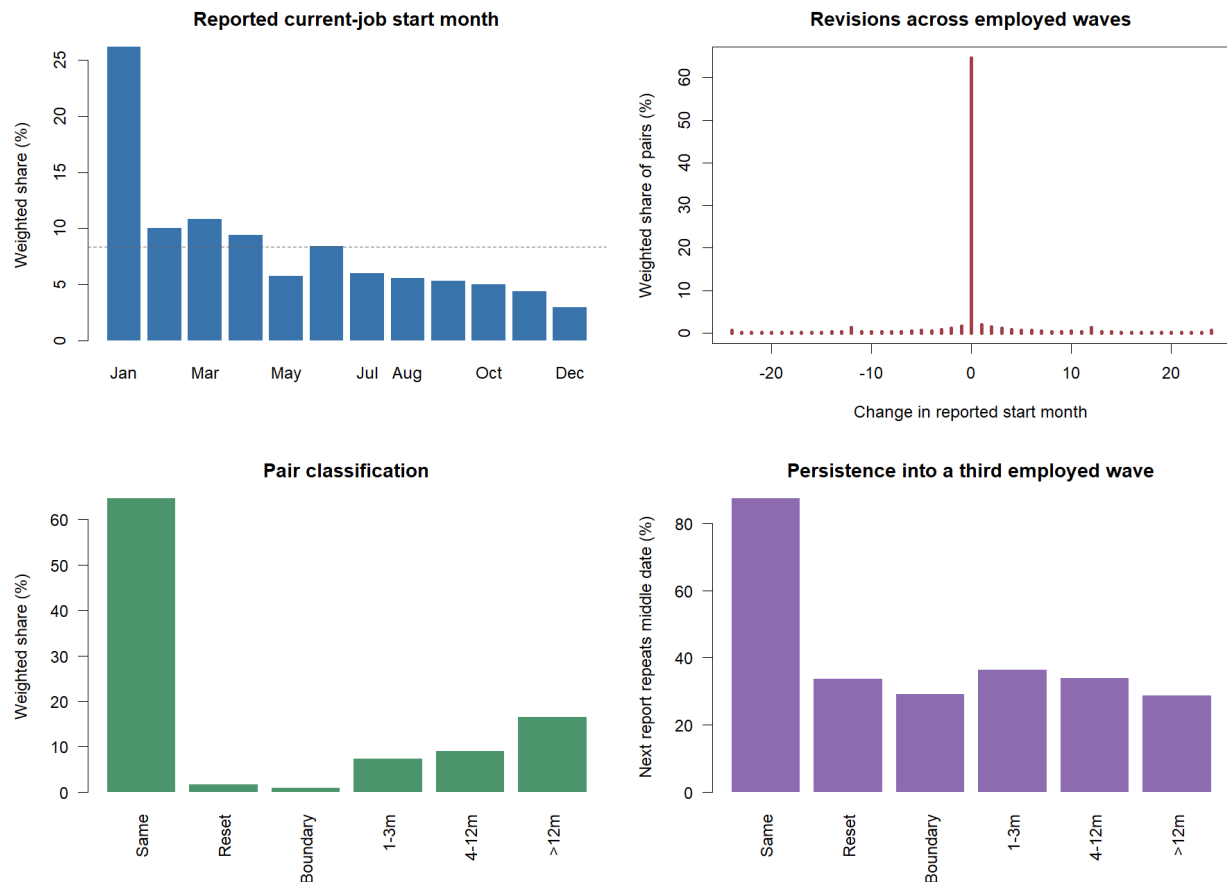


Figure 1: Reported current-job start dates and their revisions, matching panel A.

Interpretation

- Most consecutive employed reports preserve exactly the same implied start month, and an unchanged date is repeated at the following employed wave in roughly nine out of ten cases.
- Stricter matching increases the unchanged share and reduces revisions, so matching quality contributes to the observed inconsistencies.
- Large revisions remain common under panel C. They should not be attributed exclusively to either respondent reporting or record linkage.
- Exact whole-year shifts and the January concentration favor a calendar-date reporting model over a smooth local-error distribution for calculated tenure.
- A non-negligible group has a reported job start after the previous interview. Current-job tenure can therefore reset without an intervening observed nonemployment state.

3. Audit of the existing Table 7 likelihood

The preferred current specification combines piecewise duration-dependent entry and exit hazards, symmetric employment-status misclassification, gross marginal tenure contamination, and joint per-wave marginal timegap contamination. Its central duration restriction is that uncontaminated reports must follow the deterministic clock implied by a latent employment or nonemployment

spell.

For tenure, this treats a maximal latent employment spell as one current-job spell. That assumption is too strong when a worker can move directly between jobs. A legitimate job-start reset is then forced into one of three incorrect interpretations: a contaminated tenure report, a misclassified employment state, or a latent exit and re-entry.

The local-plus-gross tenure extension does not repair this problem. Its local component assigns independent bounded deviations around a single employment clock; it does not allow the underlying current-job clock itself to restart.

4. First improved model: a job-change mixture

Minimal defensible extension

For a latent employed-to-employed transition, introduce a job-continuity event (J_{it}):

$$J_{it} = 0 \quad \text{same current job,} \quad J_{it} = 1 \quad \text{new job while remaining employed.}$$

Let

$$P(J_{it} = 1 \mid H_{i,t-1} = H_{it} = 1) = q.$$

Conditional on ($J_{it}=0$), the uncontaminated tenure clock advances by the quarterly interval. Conditional on ($J_{it}=1$), the current-job tenure clock restarts within the interval. Initially, conditional event time can be taken as uniform within the quarter; a continuous-time job-change hazard can replace this approximation later.

This adds only one parameter, (q), and leaves unchanged:

- the latent employment-status transition process;
- employment-status misclassification;
- the piecewise exit and entry hazards;
- tenure-report contamination; and
- timegap contamination.

It is preferable to deterministically splitting spells whenever a date resets. The mixture allows the likelihood to weigh job change against reporting/linking inconsistency and remains coherent when reported employment status is itself misclassified.

Identification and required comparisons

The job-change probability is identified primarily by short destination tenure within a latent employed-to-employed transition and by subsequent persistence of the new start date. The first estimates must be compared on the same sample:

1. current preferred piecewise model;
2. preferred model plus (q);
3. model 2 with ($q=0$), verifying exact likelihood nesting; and
4. panels A, B, and C, verifying sensitivity to matching quality.

The estimate should not be interpreted as a pure job-to-job transition rate until identification simulations show that it can be separated from tenure contamination and status misclassification.

First estimates

The joint maximum-likelihood fit converged. It estimates a quarterly employed-to-employed current-job reset probability of 2.23%, and the survey-weighted posterior mean reset probability is 2.23%. The near equality of these two quantities is an internal consistency check.

Table 2: Preferred duration model and the one-parameter job-change extension

Model	Log likelihood	AIC	Job reset	Status error	Tenure error	Timegap error	Entry	Exit
Preferred Table 7	-2,320,295	4,640,618	0.000	0.023	0.252	0.166	0.032	0.044
Plus E-E job change	-2,316,477	4,632,985	0.022	0.025	0.237	0.166	0.030	0.043

The log likelihood rises by 3,817.5 and AIC falls by 7,633.1. A boundary likelihood-ratio statistic of 7635.1 provides overwhelming evidence against ($q=0$). Substantively, allowing legitimate current-job resets reduces the estimated gross tenure-contamination probability from 25.2% to 23.7%. It therefore explains an important subset of tenure discontinuities, but most tenure inconsistency is still assigned to the broader contamination mechanism. The weighted entry and exit rates move only modestly, so the extension improves the measurement model without overturning the employment-flow results.

Matching-panel robustness

Panel A remains the preferred and main dataset. Panels B and C change only the record-matching restrictions and diagnose whether the reset mechanism is disproportionately supported by weaker links.

Table 3: Job-change model under alternative matching restrictions

Panel	N	Job reset	Posterior reset	Status error	Tenure error	Timegap error	Entry	Exit
Panel A	402,923	0.022	0.022	0.025	0.237	0.166	0.030	0.043
Panel B	377,132	0.022	0.022	0.020	0.229	0.155	0.029	0.043
Panel C	300,200	0.019	0.018	0.014	0.191	0.135	0.028	0.042

The reset probability is nearly unchanged between Panels A and B (2.23% versus 2.19%) and declines to 1.86% in Panel C. By contrast, status misclassification falls from 2.46% to 1.37%, and

tenure contamination falls from 23.7% to 19.1%. Stricter matching therefore removes substantial measurement inconsistency but does not eliminate the current-job reset signal. The modest decline under C also cautions against interpreting every reset as a genuine job-to-job move.

5. Numerical validation and recovery

Optimizer stability

The high-reset dispersed start converges to the established optimum, and a tighter refinement changes the reset estimate only from 2.2267% to 2.2249%. A deliberately low start converges to an inferior local mode, with a log likelihood lower by 2,140.7. Multiple starts and likelihood-based selection are therefore required.

Reoptimizing all 14 nuisance parameters at seven fixed reset probabilities gives a provisional 95% profile-likelihood interval of [2.152%, 2.301%]. The local quadratic fit has ($R^2=0.99981$). This interval describes the implemented likelihood; it is not yet suitable for the paper because the recovery check below fails.

Full-likelihood recovery

Three samples of 20,000 observations were simulated from the intended model and re-estimated from both low and high reset starts. The reset and tenure-error parameters are not recovered without systematic bias.

Table 4: Full-likelihood recovery diagnostic

Parameter	Truth	Mean estimate	Bias
Job reset	0.022	0.020	-0.002
Status error	0.025	0.023	-0.001
Tenure error	0.237	0.272	0.035
Timegap error	0.166	0.164	-0.002

In particular, a true reset probability of 2.225% is estimated at 1.976% on average, while tenure contamination rises from 23.7% to 27.2%. This is not just small-sample variation: with 100,000 simulated observations, the numerical likelihood scores at the generating values remain strongly nonzero (-1713.3 for resets and 1.91752×10^4 for tenure contamination).

The leading cause is a units problem in the continuous-density approximation. Mixture branches contain different numbers of independent tenure-density evaluations, and hence have different physical units, but the current likelihood adds them directly. Clean clock restrictions also create singular components that are mixed with ordinary continuous densities. Since tenure is constructed from a reported start **month**, the defensible repair is to model tenure in discrete months: replace marginal densities by monthly interval probabilities and replace the continuous within-quarter reset density by a probability over possible start months. The job-change result should not be promoted to Table 7 until that repair restores simulation recovery.

Discrete-month repair and re-estimation

The repaired likelihood treats a reported tenure of (*m*) months as the mass

$$P(m/12 \leq D < (m+1)/12),$$

rather than evaluating a continuous density at $(m/12)$. Clean tenure clocks advance by exactly three months between quarterly interviews. A latent spell that begins between waves, or a current-job reset within a persistent employment spell, starts at 0, 1, or 2 reported months. This puts every branch of the mixture on the same probability scale. The ingestion also retains 2,753 genuine zero-month reports and drops two records that previously received an artificial quarter-month floor because no valid tenure report was available.

Table 5: Panel A discrete-month tenure likelihood

Model	Log likelihood	AIC	Job reset	Status error	Tenure error	Timegap error	Entry	Exit
No job reset	-3,178,624	6,357,276	0.000	0.031	0.223	0.165	0.023	0.041
Job reset	-3,177,000	6,354,030	0.010	0.030	0.219	0.165	0.023	0.041

The preferred reset fit estimates a quarterly current-job reset probability of 0.965%, almost exactly equal to its posterior mean of 0.965%. Allowing resets raises the log likelihood by 1,623.7 and gives a boundary LR statistic of 3247.4. Relative to the no-reset monthly model, it lowers tenure contamination only modestly, from 22.26% to 21.87%.

All three reset starts converge near the same solution: their reset estimates range from 0.960% to 0.970%, and their log likelihoods differ by less than 0.5.

Six additional full-likelihood recovery fits use three independent samples at each of 10,000 and 40,000 observations. All converge, and high dispersed starts at both sample sizes do not improve the likelihood.

Table 6: Repeated full-likelihood recovery (percentage points)

N	Parameter	Truth	Mean	Bias	RMSE
10,000	Job reset	2.227	2.346	0.120	0.132
10,000	Status error	2.459	2.459	0.001	0.109
10,000	Tenure error	23.725	23.523	-0.201	0.237
10,000	Timegap error	16.629	16.158	-0.471	0.651
40,000	Job reset	2.227	2.237	0.011	0.070
40,000	Status error	2.459	2.400	-0.058	0.083
40,000	Tenure error	23.725	23.667	-0.058	0.148
40,000	Timegap error	16.629	16.586	-0.043	0.158

At 40,000 observations, mean bias is only +0.011 percentage points for resets, -0.058 for status error, -0.058 for tenure contamination, and -0.043 for timegap contamination. RMSE also falls with sample size. The likelihood scores at the generating parameters change sign across replications rather than showing the systematic direction found under the continuous model. These checks

support accepting the discrete likelihood repair. A larger simulation would still be required to assess confidence-interval coverage directly.

Analytical and profile inference

The full 15-parameter numerical observed-information matrix is positive definite and full rank. Its minimum eigenvalue is 0.003215, and its condition number is 106.7. Delta-method standard errors therefore propagate the joint uncertainty in all transition hazards, contamination probabilities, initial employment, and job resets.

Table 7: Discrete-month model: analytical standard errors (percent)

Quantity	Estimate	SE	Lower 95%	Upper 95%
Entry	2.254	0.018	2.219	2.289
Exit	4.123	0.007	4.109	4.137
Status error	3.018	0.019	2.981	3.055
Tenure error	21.874	0.058	21.760	21.988
Timegap error	16.499	0.070	16.362	16.636
Initial employment	49.078	0.081	48.920	49.237
Job reset	0.965	0.026	0.914	1.016

The reset estimate is 0.965% with analytical SE 0.026 percentage points. Reoptimizing all 14 nuisance parameters at each fixed reset probability gives a 95% profile interval of [0.913%, 1.017%]. It is nearly identical to the Wald interval, and the profile quadratic has ($R^2=0.99987$). Thus profile asymmetry is negligible around the interior optimum.

The discrete model materially changes the substantive allocation. Compared with the continuous reset fit, the reset probability falls from 2.23% to 0.97%, status misclassification rises from 2.46% to 3.02%, and tenure contamination falls from 23.7% to 21.9%. Entry also falls, while exit changes less. Log likelihoods across the continuous and monthly models are not comparable because one uses densities and the other uses discrete probability masses.

Matching-panel sensitivity of the corrected model

The identical corrected monthly specification was re-estimated on Panels B and C. Panel A remains the main sample; the stricter panels are sensitivity checks, not alternative preferred datasets.

Table 8: Corrected discrete-month model under alternative matching restrictions (percent)

Panel	N	Job reset	Posterior reset	Status error	Tenure error	Timegap error	Entry	Exit
A	402,921	0.965	0.965	3.018	21.874	16.499	2.254	4.123
B	377,131	0.942	0.942	2.554	21.146	15.417	2.255	4.088
C	300,199	0.832	0.832	1.949	17.701	13.303	2.209	4.005

The reset estimate declines monotonically from 0.965% in Panel A to 0.942% in B and 0.832% in C. Status misclassification falls from 3.02% to 1.95%, tenure contamination from 21.9% to 17.7%, and timegap contamination from 16.5% to 13.3%. Entry is nearly unchanged in B and modestly lower in C; exit also falls slightly. Thus stricter matching removes some apparent transition and duration inconsistency, but it does not eliminate the estimated current-job reset process.

All six B/C optimizations converged normally. Within each panel, low, main, and high reset starts reach log likelihoods agreeing within 0.01 log-likelihood units; reset estimates span only 0.0035 percentage points within either Panel B or C.

Reporting-process extensions

The five nested Panel A versions can be compared coefficient by coefficient. The first rules out job resets and persistent gross reports. The second adds a job-reset probability but treats gross start-date reports as independent over time. The third estimates whether a gross report retains the preceding reported start-date anchor. The fourth instead allows gross reports to follow the empirically prominent January-heaping pattern. The fifth adds a correlated whole-year revision of the previously reported start date.

Table 9: Natural-scale coefficients across nested corrected discrete-month models (Panel A)

Coefficient	No reset	Independent gross	Persistent gross	January heap	January + year revision
Initial employment probability	0.4884	0.4908	0.4908	0.4904	0.4904
Status misclassification probability	0.0310	0.0302	0.0302	0.0304	0.0304
Tenure contamination probability	0.2226	0.2187	0.2187	0.2204	0.2202
Timegap contamination probability	0.1646	0.1650	0.1650	0.1648	0.1648
Exit hazard: 0-3 months	0.2744	0.2542	0.2542	0.2553	0.2546
Exit hazard: 3-12 months	0.2806	0.2781	0.2781	0.2783	0.2777
Exit hazard: 1-3 years	0.1665	0.1665	0.1665	0.1667	0.1668
Exit hazard: 3-5 years	0.1457	0.1457	0.1457	0.1458	0.1459
Exit hazard: 5+ years	0.1311	0.1311	0.1311	0.1312	0.1312
Entry hazard: 0-3 months	0.4478	0.4433	0.4433	0.4434	0.4433
Entry hazard: 3-12 months	0.2064	0.2047	0.2047	0.2048	0.2048
Entry hazard: 1-3 years	0.0916	0.0914	0.0914	0.0914	0.0914
Entry hazard: 3-5 years	0.1038	0.1034	0.1034	0.1034	0.1034
Entry hazard: 5+ years	0.0597	0.0543	0.0543	0.0550	0.0550
Job-reset probability	0 (fixed)	0.0097	0.0097	0.0102	0.0108
Gross-report persistence probability	0 (fixed)	0 (fixed)	0 (boundary)	0 (fixed)	0 (fixed)
January-heaping probability	0 (fixed)	0 (fixed)	0 (fixed)	0.2396	0.2353
Whole-year revision probability	0 (fixed)	0 (fixed)	0 (fixed)	0 (fixed)	0.2043
Log likelihood	- 3,178,624	-3,177,000	-3,177,000	- 3,153,877	-3,152,392
AIC	6,357,276	6,354,030	6,354,032	6,307,785	6,304,817
Number of parameters	14	15	16	16	17

All probability and hazard coefficients are shown on the natural zero-to-one scale. Exit and entry

hazards are annualized piecewise-constant hazards for the indicated duration band. The job-reset coefficient is the probability within an observed quarterly interval; gross-report persistence is the conditional probability that a contaminated report retains the previous reported anchor. January heaping is the conditional probability that a gross report is drawn from the marginal duration law restricted to an implied January start date. Whole-year revision is the conditional probability that an eligible new gross report preserves the previous reported start month but changes its year. “Fixed” denotes a nesting restriction, while “boundary” denotes an estimated coefficient whose constrained MLE is zero.

Persistent reported-start-date extension

The first reporting-process extension allows a consecutive gross tenure report within the same latent job to retain the preceding reported start-date anchor with probability (ρ). Otherwise it draws a new marginal anchor. Clean tenure reports remain exactly consistent with the true employment spell, job changes start a new segment, and the timegap process is unchanged. The independent gross-report model is nested exactly at ($\rho=0$).

Table 10: Persistent reported-start-date extension (rates in percent)

Quantity	Independent gross reports	Persistent gross anchor
Entry	2.254	2.254
Exit	4.123	4.123
Status error	3.018	3.018
Tenure error	21.874	21.874
Timegap error	16.499	16.499
Job reset	0.965	0.965
Persistence	0.000	0.000
Log likelihood	-3,177,000	-3,177,000
AIC	6,354,030	6,354,032

The constrained MLE is exactly at ($\rho=0$). Consequently, every other reported rate is unchanged, the likelihood-ratio statistic is 0.0, and adding the unused parameter raises AIC by two. The converged middle start reaches the numerical lower bound (0.000614%), while the high start returns to the same basin from ($\rho=0.90$) but reaches the iteration cap only after matching its likelihood closely. A low start finds an inferior local solution approximately 449.9 log-likelihood units below the boundary solution. The exact nested fit is therefore retained rather than a slightly inferior numerical approximation to zero.

This boundary result is not caused by an intrinsically unidentified persistence parameter. In a 12,000-record simulation with true ($\rho=0.60$), the conditional likelihood peaks at 0.550 on the reported grid, and the two full-likelihood starts converge to 0.524 and 0.524. The likelihood can therefore distinguish substantial persistence, although the best estimate is downward biased by 0.076 in this single moderate-size simulation. More replications would be needed for coverage or bias correction.

Substantively, the data do not support this particular one-parameter mechanism in addition to the clean-clock and independent gross-report branches. This is not evidence that respondents never

repeat start dates: clock-consistent sequences are already assigned to the clean branch, and a persistently wrong anchor can be observationally similar for jobs that began before the panel. The next reporting extension should therefore focus on structured start-month heaping or a persistent offset tied to jobs whose start is bounded by an observed entry or reset, rather than adding further unrestricted persistence.

January-heaped gross-report extension

The next extension uses the strongest calendar pattern in the audit. For an interview in nominal month (M), a tenure report of (m) months implies a January start when $m \bmod 12 = M - 1$. Conditional on a gross tenure report, the model now uses

$$f^{gross}(m \mid M) = (1 - h)f(m) + h \frac{\mathbf{1}\{m \bmod 12 = M - 1\}f(m)}{P_f(m \bmod 12 = M - 1)}.$$

Thus the heaped component preserves the estimated marginal duration shape but conditions it on an implied January start. It is normalized exactly, adds one parameter, and nests the independent-gross model at ($h=0$). Nominal interview months are retained when the likelihood cells are collapsed.

Table 11: January-heaped gross-report extension (rates in percent)

Quantity	Independent gross	January-heaped gross
Entry	2.254	2.263
Exit	4.123	4.128
Status error	3.018	3.043
Tenure error	21.874	22.039
Timegap error	16.499	16.484
Job reset	0.965	1.019
January heaping, conditional on gross	0.000	23.958
Gross-and-heaped tenure report	0.000	5.280
Log likelihood	-3,177,000	-3,153,877
AIC	6,354,030	6,307,785

All three starts converge normally. They estimate (h) between 0.23958 and 0.23964, while their log likelihoods differ by only 0.023. The preferred estimate is 23.96%: conditional on a gross tenure report, roughly one quarter of reports are assigned to the January-conditioned component. Because gross reporting itself has probability 22.04%, their prior product is 5.28% of otherwise correctly classified employed-wave tenure reports.

Fit improves sharply: the log likelihood rises by 23,123.6, AIC falls by 46,245.1, and the boundary likelihood-ratio statistic is 46,247.1. Entry changes from 2.254% to 2.263%, exit from 4.123% to 4.128%, and the job-reset probability from 0.965% to 1.019%. The main transition conclusions are therefore stable, while the reporting likelihood fits the calendar pattern much better.

The observed survey-weighted January share averages 26.4% across the three waves. This is not directly comparable to (h): clean reports and the ordinary marginal gross distribution can also imply January starts. Nor does the model establish that January reports are errors; genuine seasonal hiring or a coding convention could generate the same concentration. The parameter should be described as a structured reporting component unless external validation separates those mechanisms.

Table 12: Calendar-heaping recovery check (6,000 simulated records)

Parameter	Truth	Estimate	Bias
January heaping	0.4000	0.3849	-0.0151
Job reset	0.0097	0.0095	-0.0001
Status error	0.0302	0.0303	0.0001
Tenure error	0.2187	0.2139	-0.0049
Timegap error	0.1650	0.1649	-0.0001

In the recovery exercise, the truth is ($h=0.40$). The conditional optimum is 0.385, and two dispersed joint starts converge to estimates agreeing within 0.00004. The small downward error in (h) does not materially distort the reset, status, or timegap-error probabilities. This supports identification of the added component, although repeated simulations would be required for bias and coverage statements.

Table 13: Observed-information and delta-method inference for January heaping (percent)

Quantity	Estimate	Analytical SE	95% lower	95% upper
Entry	2.263	0.018	2.228	2.299
Exit	4.128	0.007	4.114	4.142
Status error	3.043	0.019	3.006	3.079
Tenure error	22.039	0.059	21.924	22.154
Timegap error	16.484	0.070	16.347	16.621
Job reset	1.019	0.026	0.967	1.070
January heaping, conditional on gross	23.958	0.143	23.677	24.239

The full 16-parameter observed-information matrix is positive definite and full rank. Its condition number is 97.2, providing no numerical indication of a locally unidentified heaping parameter. These are analytical observed-information standard errors, not bootstrap standard errors.

Correlated whole-year date revisions

The next nested component targets the second prominent calendar pattern. Let m_{t-1} be the previous gross tenure report, let $c = m_{t-1} + 3\Delta$ be its exact continuation after Δ panel intervals, and let S_{t-1} be its implied reported start month. A whole-year revision draws the current report from the marginal duration law conditional on preserving S_{t-1} while excluding (c):

$$f^{year}(m_t | S_{t-1}, c) = \frac{\mathbf{1}\{S(m_t) = S_{t-1}, m_t \neq c\}f(m_t)}{P_f\{S(m) = S_{t-1}\} - f(c)}.$$

For eligible consecutive gross reports within the same latent job, the model uses the ordinary January-heaping gross law with probability $1 - \omega$ and this different-year law with probability ω . It is normalized exactly and nests the preceding model at $\omega = 0$. This is distinct from the failed

persistence extension: persistence repeats the exact reported date, whereas this component changes the year but preserves the month.

Table 14: Correlated whole-year date-revision extension
(rates in percent)

Quantity	January heaping	January + whole-year revisions
Entry	2.263	2.264
Exit	4.128	4.126
Status error	3.043	3.041
Tenure error	22.039	22.025
Timegap error	16.484	16.485
Job reset	1.019	1.079
January heaping, conditional on gross	23.958	23.525
Whole-year revision, conditional on eligible gross pair	0.000	20.435
Log likelihood	-3,153,877	-3,152,392
AIC	6,307,785	6,304,817

The preferred high start converges normally at $\omega = 0.2043$. The central start reaches 0.2385 but is 40.0 log-likelihood points lower. The deliberately low 0.03 start stops in an inferior local mode at its initial revision probability, more than 1,011.5 points below the preferred solution. It is therefore a numerical warning, not a competing substantive estimate: multistart likelihood selection is needed. Relative to January heaping alone, the preferred log likelihood changes by 1,485, AIC changes by -2,968.1, and the boundary LR statistic is 2,970.1. Entry moves from 2.263% to 2.264%, exit from 4.128% to 4.126%, and tenure contamination from 22.039% to 22.025%.

The raw survey-weighted share of employed-report pairs whose implied start month is unchanged but whose year differs ranges from 6.68% to 7.76%. This is not directly ω : the observed share mixes clean clocks, job changes, gross reports, and status error, whereas ω is conditional on an eligible latent gross-report pair.

Table 15: Whole-year revision recovery check (6,000 simulated records)

Parameter	Truth	Estimate	Bias
Whole-year revision	0.4500	0.4581	0.0081
January heaping	0.2396	0.2391	-0.0005
Job reset	0.0102	0.0116	0.0014
Status error	0.0304	0.0325	0.0020
Tenure error	0.2204	0.2244	0.0040
Timegap error	0.1648	0.1606	-0.0042

With a generating value of $\omega = 0.45$, the conditional likelihood optimum is 0.456. Two dispersed joint starts recover 0.443 and 0.458. The best fit also recovers January heaping within 0.0005 and

tenure contamination within 0.0040 of their generating values, supporting separate identification of the two calendar components in this diagnostic sample.

Table 16: Observed-information and delta-method inference for the whole-year revision model (percent)

Quantity	Estimate	Analytical SE	95% lower	95% upper
Entry	2.264	0.018	2.229	2.299
Exit	4.126	0.007	4.112	4.141
Status error	3.041	0.019	3.004	3.078
Tenure error	22.025	0.059	21.910	22.139
Timegap error	16.485	0.070	16.348	16.622
Job reset	1.079	0.027	1.027	1.131
January heaping, conditional on gross	23.525	0.147	23.238	23.813
Whole-year revision, conditional on eligible gross pair	20.435	0.443	19.567	21.302

The full 17-parameter observed-information matrix is positive definite and full rank. Its condition number is 105.2. The analytical standard error of ω is 0.443 percentage points. The nuisance-adjusted profile interval is 19.59– 21.31%, compared with the analytical interval 19.57– 21.30%. Thus the preferred local solution has regular local curvature; this does not by itself establish that the reporting process is correctly specified.

Parametric predictive checks and source of fit improvement

The fitted-parameter checks below compare observed Panel A moments with 40 simulated samples of 50,000 records. They assess replication of the observed calendar patterns conditional on the fitted parameters; they do not integrate over parameter uncertainty.

Table 17: Fitted-parameter predictive checks for observed calendar patterns (percent)

Measure	January only	January + year revision	Observed
January start month, all employed reports	13.61	13.56	26.42
Exact continuation, consecutive employed pairs	60.32	60.32	64.70
Whole-year revision, consecutive employed pairs	3.27	4.10	7.01
Other revision, consecutive employed pairs	36.41	35.58	28.29

The revision component moves the predicted whole-year-revision share in the intended direction, from 3.27% to 4.10%, but the observed share is 7.01%. None of the four observed moments lies within the corresponding 95% simulation envelope. In particular, predicted January starts remain near 13.6% compared with 26.4% observed. A conditional January probability of about 24% among

gross reports cannot reproduce the aggregate month distribution when only about 22% of tenure reports use that gross component. Genuine hiring seasonality, calendar-specific clean reports, or both may therefore be missing from the baseline start-month law.

Table 18: Decomposition of the whole-year extension’s log-likelihood gain

Pattern	Population share (%)	Log-likelihood gain	Share of net gain (%)
Exact continuation	34.29	-70.8	-4.8
At least one whole-year revision	6.16	4568.3	307.6
Fewer than two employed reports	51.54	-7.8	-0.5
Other multiple-employed pattern	8.02	-3004.7	-202.3

All of the positive fit improvement comes from the 6.16% of weighted records with at least one observed whole-year date revision. Their gain exceeds the net gain because the extension loses about 3,083 log-likelihood points on other patterns. This concentration is substantively informative, but the offsetting deterioration and failed predictive checks show that the current correlated branch is too narrow. It requires both adjacent reports to be gross. The following extension implements the first repair; start-month seasonality remains a separate next step.

Revisions of preceding clean reports

Let η denote the probability that a gross current tenure report draws from the same normalized whole-year revision law when its preceding observed tenure report was clean. The existing gross-to-gross probability ω remains separate. Thus $\eta = 0$ reproduces the preceding 17-parameter model exactly, while the 18-parameter extension distinguishes which type of report supplies the preceding date anchor. Missing tenure reports do not erase an anchor within the same latent job, but a latent job change does.

Table 19: Clean-anchor whole-year revision extension (rates in percent)

Quantity	Gross-anchor revisions only	Gross + clean-anchor revisions
Entry	2.264	2.264
Exit	4.126	4.127
Status error	3.041	3.042
Tenure error	22.025	21.988
Timegap error	16.485	16.484
Job reset	1.079	1.085
January heaping, conditional on gross	23.525	23.845
Gross-anchor whole-year revision	20.435	12.363
Clean-anchor whole-year revision	0.000	9.641
Log likelihood	-3,152,392	-3,151,217
AIC	6,304,817	6,302,470

The extension improves the log likelihood by 1,174.7 points and lowers AIC by 2,347.4. The boundary LR statistic is 2,349.4. Starts at 3%, 25%, and 65% all converge near 9.64% for η . Allowing clean anchors reduces ω from 20.43% to 12.36%, while entry, exit, status error, and the two marginal contamination rates move by less than 0.04 percentage points. The new channel therefore reallocates the reporting process without materially changing the employment-transition interpretation.

Table 20: Observed-information inference for the clean-anchor extension (percent)

Quantity	Estimate	Analytical SE	95% lower	95% upper
Entry	2.264	0.018	2.229	2.299
Exit	4.127	0.007	4.113	4.142
Status error	3.042	0.019	3.006	3.079
Tenure error	21.988	0.059	21.873	22.102
Timegap error	16.484	0.070	16.347	16.621
Job reset	1.085	0.027	1.033	1.138
January heaping, conditional on gross	23.845	0.151	23.549	24.142
Gross-anchor whole-year revision	12.363	0.483	11.417	13.310
Clean-anchor whole-year revision	9.641	0.245	9.161	10.121

The 18-parameter observed-information matrix is positive definite and full rank, with condition number 299.5. The analytical standard error of η is 0.245 percentage points. The delta-method correlation between η and ω is -0.393. The channels are related but the local information does not indicate a nearly singular reallocation.

Table 21: Clean-anchor revision recovery check (8,000 simulated records)

Parameter	Truth	Estimate	Bias
Clean-anchor revision	0.3500	0.3247	-0.0253
Gross-anchor revision	0.2043	0.2440	0.0396
January heaping	0.2353	0.2436	0.0083
Tenure error	0.2202	0.2236	0.0033
Job reset	0.0108	0.0125	0.0017
Status error	0.0304	0.0307	0.0003

Both recovery starts converge to the same likelihood and estimate η at about 0.325 when its generating value is 0.35. The corresponding estimate of ω is about 0.244 rather than 0.204. This demonstrates joint recoverability, while the offsetting finite-sample biases caution against interpreting the two probabilities too finely without a larger repeated recovery design.

Table 22: Predictive checks after allowing clean-anchor revisions (percent)

Measure	Gross-anchor only	Gross + clean-anchor	Observed
January start month, all employed reports	13.52	13.40	26.42
Exact continuation, consecutive employed pairs	60.30	60.37	64.70
Whole-year revision, consecutive employed pairs	4.10	5.79	7.01

Measure	Gross-anchor only	Gross + clean-anchor	Observed
Other revision, consecutive employed pairs	35.60	33.84	28.29

The targeted whole-year-revision prediction rises from 4.10% to 5.79%, closer to the observed 7.01%, while other revisions fall from 35.58% to about 33.8% against 28.29% observed. The checks nevertheless still fail: exact continuations remain too low and January starts remain near 13.4% rather than 26.4%. The extension explains an important part of the date-revision pattern, but it does not repair the missing baseline calendar distribution.

Table 23: Likelihood gain from allowing clean-anchor revisions

Pattern	Population share (%)	Log-likelihood gain	Share of net gain (%)
At least one exact continuation	34.29	-1705.2	-145.2
At least one whole-year revision	6.16	3319.9	282.6
Fewer than two employed reports	51.54	-5.6	-0.5
Other multiple employed reports	8.02	-434.5	-37.0

As intended, the positive gain is concentrated among the 6.16% of weighted records with a whole-year revision. Their gain is partly offset by lower fit for exact continuations and other multiple-employed patterns. The next model should therefore estimate a genuine start-month distribution shared by clean and gross reports before adding further dependence or AR(2) dynamics.

Flexible baseline start-month distribution

The next extension replaces the implicit uniform distribution of job-start months with 12 probabilities, estimated through 11 logits relative to December. The duration mass is multiplied by the probability of its implied calendar start month and renormalized separately for each interview month. The same distribution enters clean and gross marginal tenure reports and the 0–2 month support for latent entries and job resets. Uniform probabilities reproduce the preceding 18-parameter likelihood exactly. The gross-only January-heaping branch and the two whole-year revision probabilities remain separate.

Table 24: Shared flexible baseline start-month distribution
(rates in percent)

Quantity	Uniform baseline	Flexible baseline
Entry	2.264	2.276
Exit	4.127	4.140
Status error	3.042	3.020
Tenure error	21.988	21.817
Timegap error	16.484	16.468
Job reset	1.085	1.186
January heaping, conditional on gross	23.845	0.017
Gross-anchor whole-year revision	12.363	6.576
Clean-anchor whole-year revision	9.641	10.471
Log likelihood	-3,151,217	-3,108,235
AIC	6,302,470	6,216,529

The flexible distribution improves the log likelihood by 42,981.6 points and lowers AIC by 85,941.1 despite 11 additional parameters. The LR statistic for equal month probabilities is 85,963.1 on 11 degrees of freedom. Entry, exit, status error, and tenure and timegap contamination each move by less than 0.2 percentage points. The transition interpretation is therefore stable. In contrast, the gross-specific January-heaping probability falls from 23.85% to approximately zero. January concentration belongs in the shared baseline calendar distribution rather than in an error process used only by gross reports.

Table 25: Observed and estimated job-start month distributions (percent)

Month	Observed reports	Estimated baseline	Uniform
Jan	26.42	26.46	8.33
Feb	10.03	10.28	8.33
Mar	10.89	11.20	8.33
Apr	9.49	9.66	8.33
May	5.76	5.88	8.33
Jun	8.47	8.36	8.33
Jul	6.00	6.13	8.33
Aug	5.57	5.36	8.33
Sep	5.21	5.04	8.33
Oct	4.91	4.67	8.33
Nov	4.29	4.11	8.33
Dec	2.95	2.84	8.33

The estimated baseline puts 26.46% of start dates in January, followed by March, February, and April. December has only 2.84%. This strongly non-uniform pattern explains why the earlier gross-only heaping component could not reproduce the aggregate data: most January concentration is common to clean and gross reports.

Table 26: Predictive checks for the flexible baseline (percent)

Measure	Predicted	Observed	Within 95% simulation envelope
Exact continuation	60.66	64.70	No
Whole-year revision	6.89	7.01	Yes
Other revision	32.45	28.29	No
January start month	25.85	26.42	No

The month distribution is repaired: 11 of 12 observed month shares lie inside their fitted-parameter simulation envelopes. Whole-year revisions rise to 6.89%, close to 7.01% observed. However, exact continuations remain too low (60.7% versus 64.7%) and other revisions too high (32.5% versus 28.3%). The remaining misspecification is therefore serial dependence in reporting, not baseline calendar seasonality.

Table 27: Joint recovery check (2,000 simulated records)

Quantity	Truth	Estimate
January probability	0.2646	0.2683
Month-distribution RMSE	–	0.0044
Gross-anchor revision	0.0658	0.0571
Clean-anchor revision	0.1047	0.1475
Tenure error	0.2182	0.2195
Status error	0.0302	0.0300

Uniform and reversed conditional starts converge to essentially the same likelihood, and the safeguarded joint optimizer converges before a capped central refinement. The 12-month distribution is recovered with RMSE about 0.0044. The two revision channels show more small-sample reallocation, so their sum and predictive implications remain more reliable than a fine comparison of their separate probabilities. Full observed-information inference for all 29 parameters remains to be completed; the near-zero gross-only January-heaping estimate is a boundary quantity and should not be given a regular Wald interpretation.

Exact reported-start-date retention

The next nested extension lets a gross current tenure report retain the preceding reported job-start date exactly, regardless of whether the preceding report was clean or gross. Let ρ denote this retention probability. If the anchor is not retained, the report follows the existing whole-year revision versus marginal-redraw mixture. Thus $\rho = 0$ reproduces the flexible baseline likelihood exactly and the two revision parameters retain their conditional interpretation.

Table 28: Exact reported-start-date retention (rates in percent)

Quantity	Flexible baseline	Exact retention
Entry	2.276	2.308
Exit	4.140	4.142
Status error	3.020	2.978
Tenure error	21.817	25.169
Timegap error	16.468	16.494
Job reset	1.186	1.245
Gross-anchor whole-year revision	6.576	6.599
Clean-anchor whole-year revision	10.471	12.254
Exact-anchor retention	0.000	19.970
Log likelihood	-3,108,235	-3,107,540
AIC	6,216,529	6,215,139

The polished joint estimate is 19.97%. The log likelihood rises by 695.8 points; the corresponding one-parameter LR statistic is 1,391.6. Entry and exit rates move only from 2.28% and 4.14% to 2.31% and 4.14%. The substantive transition result is therefore stable. Tenure contamination rises from 21.82% to 25.17% because the enlarged gross-report process can now include both retained and redrawn dates.

Table 29: Longer refinement of the exact-retention model

Fit	Log likelihood	Gain	Retention (%)
First central block	-3,107,542	+0.00	19.45
Low-retention start	-3,108,094	-552.23	3.05
High-retention start	-3,107,936	-394.24	32.05
Continued joint fit	-3,107,540	+2.02	19.96
Conditional polish	-3,107,540	+2.02	19.97

The continued joint refinement raises the log likelihood by 2.02 points relative to the first central block. Low- and high-retention screens are hundreds of likelihood points worse. The final scalar polish adds 0.001 point, so the point estimates are practically stable even though the capped joint blocks retain optimizer convergence code 1. Full observed-information inference for the 30 parameters remains outstanding.

Table 30: Predictive implications of exact-anchor retention (percent)

Measure	Model	Predicted	Observed
Exact continuation	Exact-anchor retention	61.79	64.70
Exact continuation	Flexible baseline	60.64	64.70
Whole-year revision	Exact-anchor retention	6.80	7.01
Whole-year revision	Flexible baseline	6.93	7.01
Other revision	Exact-anchor retention	31.40	28.29
Other revision	Flexible baseline	32.43	28.29

Exact continuations rise from 60.6% to 61.8%, toward the observed 64.7%, and other revisions fall from 32.4% to 31.4%, toward 28.3%. Whole-year revisions remain essentially exact at 7.0%. The extension therefore targets the intended serial feature but does not close the full gap. Likelihood gains are concentrated in records containing a mixture of continuation patterns and are offset by losses among single-pattern records. A richer local date-revision kernel remains warranted.

Table 31: Conditional recovery of exact-anchor retention (20 samples of 5,000)

Quantity	Value
Generating retention	0.1997
Mean recovered retention	0.1971
Recovery RMSE	0.0283
Minimum estimate	0.1489
Maximum estimate	0.2451
Mean LR against zero	60.4682

Holding nuisance parameters at their generating values, the new probability is recovered with small mean bias and an RMSE of about three percentage points. All 20 replications produce a positive interior estimate and strongly favor retention over the nested zero value. This establishes conditional identification, but is intentionally an easier check than joint recovery.

Table 32: Full 30-parameter recovery: transition rates (percent)

Sample	True entry	Estimated entry	True exit	Estimated exit
Replication 1	2.534	2.548	4.201	4.208
Replication 2	2.564	2.542	4.216	4.313
Replication 3	2.616	2.823	4.220	4.374

Table 33: Full 30-parameter recovery: reporting probabilities (percentage points)

Parameter	Truth	Mean	Bias	RMSE
Exact-anchor retention	19.97	20.64	+0.67	4.68
Tenure contamination	25.17	24.95	-0.22	1.46
Status misclassification	2.98	3.02	+0.04	0.19
Clean-anchor revision	12.25	14.26	+2.00	2.57
Gross-anchor revision	6.60	5.46	-1.14	2.85

All three simulated samples can be re-estimated from near and dispersed starts. The selected pooled entry- and exit-rate RMSEs are 0.121 and 0.105 percentage points. The third replication is less favorable than the first two, with entry and exit errors of +0.207 and +0.154 points, respectively. The aggregate transition result is nevertheless much more robust than individual duration-hazard bins. By contrast, the reporting decomposition is weakly separated at this sample size: the retention RMSE is 4.68 points, and clean- and gross-anchor revision probabilities partly offset one another. The exercise shows that the full likelihood is estimable, but three replications do not establish precise finite-sample identification of every reporting component.

Table 34: OPG analytical inference for the exact-retention model (percent)

Quantity	Estimate	Analytical SE	95% interval
Entry	2.308	0.019	[2.272, 2.345]
Exit	4.142	0.006	[4.130, 4.155]
Status error	2.978	0.020	[2.939, 3.016]
Tenure error	25.169	0.132	[24.911, 25.427]
Timegap error	16.494	0.074	[16.348, 16.640]
Job reset	1.245	0.029	[1.189, 1.301]
Gross-only January heaping	0.017	0.289	[-0.549, 0.583]
Gross-anchor revision	6.599	0.497	[5.625, 7.573]
Clean-anchor revision	12.254	0.324	[11.618, 12.890]
Exact-anchor retention	19.970	0.574	[18.846, 21.095]

The 30-parameter score matrix is full rank, but its condition number is $5.35\text{e}+07$, consistent with weak local separation. The exact-retention standard error is 0.574 percentage points under this analytical approximation. The most important natural-scale correlations are 0.885 between exact retention and overall tenure contamination, and -0.540 between the clean- and gross-anchor revision probabilities. These correlations agree with the simulation evidence that reporting channels can offset one another. Gross-only January heaping is effectively on the boundary: its regular interval crosses zero by a wide margin, so its Wald standard error should be treated only as a diagnostic. The OPG calculation uses 61 likelihood evaluations. A resumable central observed-Hessian calculation has also been implemented; its first 48 of 1,801 points are cached, but the exact calculation is not yet complete.

Local reported-start-date revisions

The next nested extension addresses near-anchor changes directly. Conditional on a gross current tenure report, the preceding reported start date is first retained exactly with probability ρ . If it is not retained, the report receives a non-zero local revision with probability κ ; otherwise it passes to the existing whole-year-revision versus marginal-redraw mixture. The local revision is a normalized discrete-Laplace kernel over one to six months in either direction, with fixed three-month scale and truncation at zero tenure. Consequently, $\kappa = 0$ reproduces the exact-anchor model exactly and zero-month changes are not counted twice.

Table 35: Local reported-start-date revision extension (rates in percent)

Quantity	Exact retention	Local revision
Entry	2.308	2.308
Exit	4.142	4.128
Status error	2.978	2.960
Tenure error	25.169	25.659
Timegap error	16.494	16.502
Job reset	1.245	1.207
Gross-anchor whole-year revision	6.599	6.990
Clean-anchor whole-year revision	12.254	14.986
Exact-anchor retention	19.970	22.098
Local revision conditional on no exact retention	0.000	33.845
Log likelihood	-3,107,540	-3,078,029
AIC	6,215,139	6,156,119

The conditional local-revision estimate is 33.85%. Because exact retention is tried first, its corresponding unconditional share among eligible gross reports is 26.37%. The log likelihood increases by 29,510.9 points, giving a boundary LR statistic of 59,021.8 and a much lower AIC. The conditional profile peaks near 0.35. Reporting-parameter fits started near 0.02, the conditional optimum, and 0.65; the low and high screens are respectively 1,503.7 and 4,360.1 likelihood points below the best result. A central-difference reporting block improves the main forward result and a separate transition- hazard block adds only 4.61 points. The blocks are deliberately short and retain convergence code 1, so this is strong multistart evidence but not a claim of final full convergence.

Entry changes from 2.308% to 2.308% and exit from 4.142% to 4.128%. Status error also remains near 2.96%. The new channel therefore changes the measurement process while leaving the paper's transition interpretation essentially intact.

Table 36: Predictive implications of local anchor revisions
(percent)

Measure	Model	Predicted	Observed
Exact continuation	Exact-anchor model	61.66	64.70
Exact continuation	Local anchor revisions	61.74	64.70
Local revision (1 to 6 months)	Exact-anchor model	3.04	11.43
Local revision (1 to 6 months)	Local anchor revisions	10.79	11.43
Whole-year revision	Exact-anchor model	6.82	7.01
Whole-year revision	Local anchor revisions	5.26	7.01
Other revision	Exact-anchor model	28.49	16.87
Other revision	Local anchor revisions	22.20	16.87

The old exact-anchor model predicts only 3.0% local revisions, compared with 11.4% observed. The extension raises the prediction to 10.8%. Local-only records contribute 21,119 likelihood points and mixed-pattern records contribute another 18,281; losses on other and whole-year-only records partly offset these gains. This confirms that the extension improves the feature it was designed to capture rather than obtaining its fit from unrelated observations.

The check is not yet fully satisfactory. Exact continuations remain underpredicted (about 61.7% versus 64.7%), local revisions remain slightly underpredicted (10.8% versus 11.4%), and residual revisions remain overpredicted (22.2% versus 16.9%). Whole-year revisions now fall below the observed share. The fixed local kernel is therefore a useful first repair, not a complete reporting model. Holding all other parameters fixed, the local probability has a curvature-based standard error of 0.215 percentage points and a 95% interval of [33.42%, 34.27%]. This is conditional rather than full nuisance-adjusted inference and should not be used as the final paper standard error.

Shared person-level duration-reporting reliability

The next extension allows tenure and timegap reports from the same person to share a latent propensity for contamination. Each person belongs with equal prior probability to a reliable or unreliable reporting class. Relative to a common midpoint, the reliable class shifts both contamination logits down by δ and the unreliable class shifts them up by δ . Thus $\delta = 0$ reproduces the preceding independent-reporting likelihood exactly. The fixed equal class shares keep this first extension parsimonious and avoid an additional mixing-proportion parameter and label switching.

Table 37: Shared duration-reporting reliability (rates in percent)

Quantity	Independent	Shared reliability
Entry	2.308	2.311
Exit	4.128	4.128
Status error	2.960	2.960
Job reset	1.207	1.207
Exact-anchor retention	22.098	22.098
Local revision	33.845	33.845
Reliability-logit dispersion	0.000	1.688
Log likelihood	-3,078,029	-3,064,447
AIC	6,156,119	6,128,958

The likelihood rises by 13,581.4 points with one additional parameter, while entry remains 2.311% and exit remains 4.128%. The reporting extension therefore leaves the substantive transition estimates essentially unchanged. The exact equality of the two likelihoods at $\delta = 0$ was verified numerically. The LR statistic is 27,162.9, but its usual one-degree-of-freedom chi-squared reference distribution is not valid because the null is a homogeneous-mixture boundary. Final inference requires a parametric bootstrap.

Table 38: Implied duration-contamination probabilities by latent reliability class (percent)

Class	Prior share	Tenure contamination	Timegap contamination
Reliable	50.0	4.36	1.73
Unreliable	50.0	57.11	33.91
Equal-prior marginal mean	—	30.73	17.82

The reliable class has tenure and timegap contamination probabilities of 4.36% and 1.73%, compared with 57.11% and 33.91% in the unreliable class. The survey-weighted mean posterior probability of the unreliable class is 48.7%, with a median of 37.8%. The midpoint contamination parameters reported by the optimizer are not population-average probabilities in this mixture; the class-specific values and equal-prior marginal means above are the interpretable quantities.

All three deliberately dispersed joint starts improve on the independent model. The best low and high starts remain respectively 8,504.5 and 1,750.3 likelihood points below the preferred basin. The

short joint blocks terminate at their iteration caps, however, so the result is a strong provisional point estimate rather than a final convergence claim.

Table 39: Predictive implications of shared reporting reliability (percent)

Measure	Observed	Independent	Shared reliability
Exact continuation	64.70	61.66	61.86
Local revision (1 to 6 months)	11.43	10.77	11.13
Other revision	16.87	22.26	22.06
Timegap inconsistency	15.48	17.97	17.82
Two opportunities: no irregularity	68.05	60.36	62.88
Two opportunities: one irregularity	16.43	24.07	19.39
Two opportunities: two irregularities	15.52	15.58	17.73

Shared reliability improves the local-revision prediction from 10.77% to 11.13% against 11.43% observed and moves the repeated-opportunity distribution in the right direction: the predicted shares with no and exactly one irregularity change from 60.36% and 24.07% to 62.88% and 19.39%, against 68.05% and 16.43% observed. It is not a complete repair. The model raises the two-irregularity share from 15.58% to 17.73% against 15.52% observed, and the exact, residual-revision, and timegap moments remain outside their simulation intervals.

The likelihood decomposition confirms that the improvement is concentrated where duration information is informative: records with no irregularity, tenure deviations, and timegap inconsistencies account for 65.6%, 28.9%, and 5.8% of the gain, respectively. The average posterior unreliable probability is 28.0% for regular repeated-state records, 87.5% for tenure deviations, and 92.6% for timegap inconsistencies.

There is an important identification limitation. Across only three waves, a record cannot contain both a repeated employed spell and a repeated nonemployment spell. The common shift across tenure and timegap is therefore not identified from observing both clocks repeatedly for the same person. Instead, it is a parsimonious cross-clock restriction learned indirectly from serial clustering within each clock. The following specification check allows separate tenure- and timegap-reliability dispersions and tests whether the common-shift restriction is supported.

Relaxing the common-shift restriction The restriction test replaces the single δ with separate nonnegative tenure and timegap dispersions, while retaining the same person-level class and equal prior shares. Setting the two dispersions equal reproduces the common-shift likelihood exactly; this was verified to numerical precision.

Table 40: Relaxing equal tenure and timegap reliability dispersions (rates in percent)

Quantity	Common dispersion	Separate dispersions
Entry	2.385	2.385
Exit	4.146	4.146
Status error	2.856	2.856
Job reset	0.963	0.963
Exact-anchor retention	25.248	25.248
Local revision	33.312	33.312
Tenure dispersion	2.161	2.088
Timegap dispersion	2.161	2.216
Log likelihood	-3,063,169	-3,063,097
AIC	6,126,402	6,126,259

The estimated tenure dispersion is 2.088, compared with 2.216 for timegap. The log likelihood rises by 72.3 points and AIC falls by 142.6. The equality-restriction LR statistic is 144.6 on one degree of freedom. This is a like-for-like comparison: the unrestricted fit was continued jointly until a further step made no improvement, the restricted fit was initialized by projecting those nuisance estimates onto a common shift, its common shift was polished conditionally, and a final full restricted step also made no improvement. Both retained fits have convergence code zero. The earlier unrefined LR statistic was much larger because it incorrectly attributed unrelated nuisance-parameter refinement to the single dispersion restriction.

Entry and exit remain 2.385% and 4.146%. Relaxing the restriction therefore changes the duration-reporting decomposition without changing the paper’s substantive employment-transition estimates.

Table 41: Contamination probabilities under separate reliability dispersions (percent)

Class	Prior share	Tenure contamination	Timegap contamination
Reliable	50.0	2.18	0.59
Unreliable	50.0	59.24	33.27
Equal-prior marginal mean	–	30.71	16.93

The unrestricted model sharpens the tenure separation: tenure contamination is 2.18% in the reliable class and 59.24% in the unreliable class. The corresponding timegap probabilities are 0.59% and 33.27%. The survey-weighted mean posterior unreliable probability is 49.9%.

Table 42: Provisional full-nuisance OPG inference (rates in percent)

Quantity	Estimate	OPG standard error
Entry rate	2.385	(0.020)
Exit rate	4.146	(0.007)
Status error	2.856	(0.020)
Tenure contamination: reliable	2.185	(0.084)
Tenure contamination: unreliable	59.240	(0.224)
Timegap contamination: reliable	0.590	(0.125)
Timegap contamination: unreliable	33.272	(0.199)
Tenure dispersion	2.088	(0.020)
Timegap dispersion	2.216	(0.108)
Timegap minus tenure dispersion	0.128	(0.110)

The 33-parameter OPG matrix is full rank and positive definite, but its condition number is 55,368,340. The fourth exit-hazard perturbation required a tenfold smaller finite-difference step. Most importantly, the OPG Wald interval for the difference between the two dispersions includes zero even though the directly optimized LR statistic is large. The OPG is an outer-product approximation rather than the exact observed Hessian, and the mixture likelihood is evidently not well approximated by one common local quadratic for this contrast. The LR comparison is therefore the preferred evidence about the equality restriction; the OPG errors are useful provisional uncertainty measures for the headline rates but not a substitute for a profile or bootstrap of the dispersion difference.

Table 43: Nuisance-adjusted profile inference for the dispersion difference

Quantity	Result
Timegap minus tenure dispersion	0.1281
95% profile interval	[0.1100, 0.1520]
LR test of equal dispersions	144.65
LR p-value	2.57e-33

The direct profile fixes the timegap-minus-tenure dispersion difference and reoptimizes the other 32 parameters. Each outer bracket was screened from a nearby solution and from the common-dispersion nuisance estimates; all retained fits converged, and the nearby starts gave the higher likelihood on both sides. Tightly optimized points near both confidence limits give a 95% profile interval from 0.1100 to 0.1520, which excludes zero by a wide margin. The profile therefore agrees with the fully optimized LR test and shows that the broad OPG Wald interval is a numerical/local-quadratic failure for this specific contrast. It does not invalidate the provisional OPG uncertainty measures for the headline transition rates.

A 10,000-person simulation-recovery check held the remaining nuisance parameters at their generating values and estimated the two midpoints and two dispersions from three widely separated starts. All starts converged to the same likelihood. Tenure midpoint and dispersion biases were

-0.491 percentage points and +0.048; timegap was noisier, with corresponding biases of -1.105 points and +0.256. The posterior reliability score classified the simulated latent groups with AUC 0.871. This is encouraging but remains a one-replication, conditional recovery diagnostic rather than a complete finite-sample validation of all 33 parameters.

Table 44: Predictive implications of separate reliability dispersions (percent)

Measure	Observed	Common dispersion	Separate dispersions
Exact continuation	64.70	63.76	64.12
Local revision (1 to 6 months)	11.43	10.56	10.36
Other revision	16.87	20.35	20.23
Timegap inconsistency	15.48	16.60	17.07
Two opportunities: no irregularity	68.05	65.03	64.83
Two opportunities: one irregularity	16.43	18.18	18.43
Two opportunities: two irregularities	15.52	16.79	16.74

Separate dispersions improve exact continuations from 63.8% to 64.1% against 64.7% observed. They change the predicted share with two regular duration opportunities from 65.0% to 64.8% against 68.1%, and increase predicted timegap inconsistencies from 16.6% to 17.1% against 15.5%. The local-revision prediction also moves away from its observed value, and all reported moments remain outside the simulation intervals. The extension is preferred by the observed likelihood, but is not an adequate reporting model on these coarse predictive checks.

The likelihood decomposition makes the tradeoff explicit. Regular repeated-state records lose about 591 likelihood points relative to the fully refined common-dispersion model, tenure deviations lose about 559, and records without repeated-state duration information lose about 9. These are more than offset by a gain of about 1,231 among timegap-inconsistent records. Thus the likelihood improvement comes from fitting the distribution within the timegap-inconsistent group, not from reducing its aggregate predicted frequency.

6. Script and output register

diagnostics/audit_job_start_dates.R. Implemented. Reconstructs reported start months and audits heaping, revisions, persistence, and matching sensitivity. It is a base-R script, accepts an A/B/C panel path, and writes panel-specific and combined outputs.

EM-tenure/R/job_change_eps.R. Implemented. Enumerates same-job and new-job branches inside full latent employment spells, exactly nests the old emission at ($q=0$), uses log-sum-exp, and exposes posterior job-change counts.

EM-tenure/estimate_job_change_tenure_contamination.R. Implemented. Profiles (q), jointly estimates it with all nuisance parameters, rejects an invalid or inferior fit, and preserves the preferred Table 7 fit as the nested comparison. It does not bootstrap.

EM-tenure/estimate_job_change_matching_robustness.R. Implemented. Re-estimates the identical 15-parameter model on Panels B and C from the Panel A optimum and writes separate robustness outputs without replacing the main fit.

EM-tenure/validate_job_change_model.R. Implemented. Runs dispersed starts, tight refinement, a nuisance-adjusted profile, multistart full-model recovery simulations, and a large-sample score diagnostic. The recovery check currently fails and identifies the monthly-mass repair as the next priority.

EM-tenure/R/tenure_monthly_eps.R. Implemented. Replaces continuous tenure densities with monthly probability masses, advances clean clocks by three months, and uses a discrete 0–2 month support for entries and resets.

EM-tenure/validate_monthly_tenure_model.R. Implemented. Checks mass normalization, evaluates large-sample scores, and re-estimates simulated data from dispersed starts. The first full-likelihood recovery exercise is close to the generating parameters.

EM-tenure/validate_monthly_tenure_recovery_grid.R. Implemented. Runs a resumable recovery grid at multiple sample sizes, repeats dispersed-start checks, and reports bias, RMSE, hazard recovery, and likelihood scores.

EM-tenure/estimate_monthly_job_change_tenure_contamination.R. Implemented. Preserves reported zero-month tenure, estimates the no-reset and reset models on Panel A, and compares three reset starting values.

EM-tenure/estimate_monthly_job_change_matching_robustness.R. Implemented. Re-estimates the corrected monthly model on Panels B and C, checks the monthly grid, screens low, main, and high reset starts, and writes separate cached robustness outputs without replacing Panel A.

EM-tenure/estimate_monthly_persistent_tenure_reporting.R. Implemented. Adds a within-job retain-versus-redraw process for consecutive gross tenure reports, checks exact nesting, screens three persistence starts, and retains the exact boundary fit when all positive approximations are inferior.

EM-tenure/validate_monthly_persistent_tenure_reporting.R. Implemented. Simulates positive persistence, evaluates a conditional profile, and performs two full-likelihood recovery fits from dispersed starts.

EM-tenure/estimate_monthly_calendar_heaping.R. Implemented. Retains the nominal interview quarter, adds a normalized January-conditioned component to gross tenure reports, verifies exact nesting, and screens three Panel A starts.

EM-tenure/validate_monthly_calendar_heaping.R. Implemented. Simulates a positive January-heaping probability, evaluates its conditional likelihood, and checks joint recovery from low and high starts.

EM-tenure/infer_monthly_calendar_heaping.R. Implemented. Evaluates the full numerical observed-information matrix in resumable parallel chunks and uses the delta method for the reported rates and calendar-heaping quantities.

EM-tenure/estimate_monthly_calendar_revision.R. Implemented. Adds a normalized, correlated whole-year revision branch to eligible consecutive gross reports, verifies exact nesting, and screens three Panel A starts.

EM-tenure/validate_monthly_calendar_revision.R. Implemented. Verifies conditional identification and joint recovery of whole-year revision and January-heaping probabilities from dispersed starts.

EM-tenure/infer_monthly_calendar_revision.R. Implemented. Computes the full 17-parameter observed-information matrix in resumable parallel chunks and uses the delta method for all reported rates.

EM-tenure/profile_monthly_calendar_revision.R. Implemented. Reoptimizes all nuisance parameters from warm and alternative starts on both sides of the whole-year revision estimate and interpolates the 95% profile interval.

EM-tenure/diagnose_monthly_calendar_revision.R. Implemented. Simulates calendar moments at fitted parameters and decomposes the exact observed-data likelihood gain by mutually exclusive observed patterns.

EM-tenure/estimate_monthly_clean_anchor_revision.R. Implemented. Adds a separate whole-year revision probability for a current gross report following a clean reported date, verifies exact nesting, and screens three starts.

EM-tenure/validate_monthly_clean_anchor_revision.R. Implemented. Checks the conditional likelihood and joint recovery of clean-anchor, gross-anchor, and January-heaping probabilities from two dispersed starts.

EM-tenure/diagnose_monthly_clean_anchor_revision.R. Implemented. Compares fitted calendar moments with the nested model and decomposes the row-level likelihood gain by observable report patterns.

EM-tenure/estimate_monthly_start_month_baseline.R. Implemented. Adds 11 month logits shared by clean and gross marginal reports, verifies exact nesting at equal probabilities, screens three conditional starts, and performs staged joint Panel A refinement.

EM-tenure/refine_monthly_start_month_baseline.R. Implemented. Continues the full-sample central-difference refinement without repeating the conditional mode search.

EM-tenure/diagnose_monthly_start_month_baseline.R. Implemented. Compares the full fitted month distribution and consecutive-report patterns with the nested model and decomposes the likelihood gain by observed start month.

EM-tenure/validate_monthly_start_month_baseline.R. Implemented. Checks distant month-distribution starts and joint recovery, including a safeguarded one-sided gradient near the zero-heaping boundary.

EM-tenure/estimate_monthly_exact_anchor_retention.R. Implemented. Adds a single retain-versus-revise/redraw channel after either a clean or gross reported anchor, verifies exact nesting, profiles the new parameter widely, and performs staged joint refinement plus a conditional polish.

EM-tenure/diagnose_monthly_exact_anchor_retention.R. Implemented. Compares exact, whole-year, and other consecutive-report changes with fitted simulations and decomposes the observed likelihood gain by report pattern.

EM-tenure/validate_monthly_exact_anchor_retention.R. Implemented. Runs 20 conditional recovery samples at the empirical generating value and records bias, RMSE, range, and the LR statistic against zero.

EM-tenure/refine_monthly_exact_anchor_retention.R. Implemented. Resumes the saved 30-parameter fit, compares continued, low-retention, and high-retention starts, promotes the best joint fit, and finishes with an exact conditional polish without discarding cached work.

EM-tenure/validate_monthly_exact_anchor_joint_recovery.R. Implemented. Simulates the complete model and re-estimates all 30 parameters from near and dispersed starts, with resumable replication caches and separate summaries of transition, hazard, calendar-baseline, and reporting-process recovery.

EM-tenure/infer_monthly_exact_anchor_retention_opg.R. Implemented. Uses 61 finite-difference likelihood evaluations and record-level scores to obtain a full-rank OPG information approximation, delta-method errors, boundary flags, and correlations among natural-scale reporting probabilities.

EM-tenure/infer_monthly_exact_anchor_retention.R. Implemented as a resumable exact central observed-Hessian calculation. It requires 1,801 full likelihood evaluations; 48 are currently cached, so this remains a confirmatory calculation rather than the reported inference result.

EM-tenure/estimate_monthly_local_anchor_revision.R. Implemented. Adds a normalized one-to-six-month revision kernel conditional on not retaining the preceding anchor exactly. It verifies exact nesting, profiles the new probability, screens low and high starts, refines the competing reporting parameters, checks the transition-hazard block, and caches every completed stage. The calendar fitter now accepts named free-parameter blocks while retaining its full-fit default.

EM-tenure/diagnose_monthly_local_anchor_revision.R. Implemented. Compares exact, local, whole-year, and residual revisions in observed and simulated data, decomposes the likelihood gain by report pattern, and reports a clearly labelled conditional-curvature approximation for the new probability.

EM-tenure/estimate_monthly_shared_duration_reliability.R. Implemented. Adds an equal-weight person-level reliability mixture shared by tenure and timegap contamination, verifies exact nesting at zero dispersion, profiles the new parameter, and performs resumable dispersed-start joint refinement.

EM-tenure/diagnose_monthly_shared_duration_reliability.R. Implemented. Compares observed and fitted duration-irregularity patterns, decomposes the row-level likelihood gain, and summarizes posterior unreliable-class probabilities without interpreting class membership as an observed trait.

EM-tenure/estimate_monthly_separate_duration_reliability.R. Implemented. Allows tenure and timegap contamination to have separate reliability dispersions, verifies exact nesting when they are equal, maps the conditional surface, and screens common and asymmetric starts before a conditional polish.

EM-tenure/diagnose_monthly_separate_duration_reliability.R. Implemented. Compares fitted duration-pattern distributions under common and separate dispersions and decomposes their row-level likelihood difference.

Separate-reliability refinements. Implemented in:

EM-tenure/refine_monthly_separate_duration_reliability.R

EM-tenure/refine_monthly_separate_reliability_blocks.R

EM-tenure/refine_monthly_separate_reliability_joint.R

Continue the four focal parameters, test reporting, transition-hazard, and full blocks, and finish in interruption-safe joint checkpoints.

Common-reliability refinements. Implemented in:

EM-tenure/refine_monthly_common_duration_reliability.R

EM-tenure/polish_monthly_common_duration_reliability.R

Project the converged nuisance estimates into the equality-restricted model, polish its common shift, and verify that a full restricted step cannot improve the comparison likelihood.

EM-tenure/validate_monthly_separate_duration_reliability.R. Implemented. Runs cached multiple-start conditional recovery simulations for the two midpoints and two dispersions and checks posterior latent-class separation.

EM-tenure/infer_monthly_separate_duration_reliability_opg.R. Implemented. Computes full-nuisance OPG/delta-method errors from cached record-level scores, with adaptive perturbations and explicit rank, conditioning, and score diagnostics.

Separate-reliability profile. Implemented in:

EM-tenure/profile_monthly_separate_duration_reliability.R

EM-tenure/refine_monthly_separate_duration_reliability_profile_limits.R

Reparameterize the clock-specific dispersions by their average and fixed difference, screen both profile sides from distinct nuisance starts, and tightly refine the two 95% likelihood-ratio crossings.

Analytical-inference script. Implemented. Computes the full observed-information matrix in parallel, propagates it to reported rates by the delta method, and reoptimizes all nuisance parameters for the reset-probability profile.

Profile-refinement script. Implemented. Refines a single fixed reset-probability point from the nearest converged saved profile fit.

EM-tenure/tests/testthat/test-job-change-eps.R. Implemented. Covers unit behavior, exact emission and complete-likelihood nesting, finite likelihoods, strict validation, posterior reset classification, and recovery of (q) in a simulated emission model.

Generated diagnostics belong under `diagnostics/output/job_start_dates/`; fitted-model outputs belong under `EM-tenure/output/results/job_change/`. Both are ignored working artifacts.

7. Acceptance checks for the first extension

The first empirical result should not be added to Table 7 until all of the following are complete. Checks already passed are marked accordingly:

- exact equality to the existing likelihood at (q=0) — **passed**;
- finite likelihoods at interior (q) values and valid posterior probabilities — **passed**;
- optimizer convergence from the nested fit — **passed**;
- no deterioration relative to the nested model — **passed**;
- recovery of (q) in a simplified emission simulation — **passed**;
- dispersed starts and tighter tolerances — **passed**, with an inferior low-reset local mode that makes multistart selection necessary;
- recovery of (q) and nuisance parameters in full-likelihood simulations — **failed** under the current continuous-density approximation;
- discrete monthly masses sum to one and zero-month reports are retained — **passed**;

- discrete-month full-likelihood recovery from dispersed starts — **passed** in six repeated fits at 10,000 and 40,000 observations, with bias and RMSE contracting at the larger sample;
- discrete-month Panel A convergence from low, main, and high reset starts — **passed**, with closely agreeing estimates and log likelihoods;
- matching-panel sensitivity — **passed** for both the continuous diagnostic and the corrected discrete model; all B/C starts converge, while reset and contamination estimates decline monotonically with stricter matching; and
- full-rank positive-definite observed information and delta-method errors — **passed**; and
- nuisance-adjusted profile-likelihood inference — **passed**, with all retained points converged and a profile interval nearly identical to the analytical interval; and
- positive persistence recoverability — **passed provisionally** in one 12,000-record simulation; both joint starts identify substantial positive persistence, although additional replications are required to characterize its finite-sample bias; and
- January-heaping normalization, nesting, recovery, and Panel A multistart convergence — **passed**; all three empirical starts estimate about 24% conditional heaping, and a 6,000-record simulation closely recovers a 40% generating probability; and
- January-heaping observed-information inference — **passed**; the full 16-parameter information matrix is positive definite and full rank; and
- whole-year revision normalization, nesting, recovery, and Panel A multistart convergence — **passed**; the recovery exercise separates the revision and January-heaping components, and the empirical results are reported above;
- whole-year revision observed-information and nuisance-profile inference — **passed**; the information matrix is full rank and positive definite, and the two interval estimates agree closely; and
- whole-year revision predictive adequacy — **failed**; the extension moves the targeted moment in the correct direction but materially underpredicts both January starts and whole-year revisions, while its positive likelihood gain is concentrated in 6.16% of weighted records;
- clean-anchor revision nesting, multistart convergence, and joint recovery — **passed provisionally**; three empirical starts agree and two simulated starts recover both revision channels, although an 8,000-record exercise shows offsetting finite-sample bias between them;
- clean-anchor observed-information inference — **passed**; the full 18-parameter matrix is positive definite and full rank; and
- clean-anchor predictive adequacy — **failed but improved**; the predicted whole-year revision share moves from 4.10% to 5.79% against 7.01% observed, while January starts and exact continuations remain poorly reproduced;
- shared baseline start-month normalization and exact nesting — **passed**; equal probabilities reproduce the 18-parameter model exactly and the calendar-weighted duration masses sum to one for every interview month;
- baseline start-month multistart and recovery — **passed provisionally**; three empirical conditional starts and distant simulated starts meet, and a 2,000-record joint recovery gives month-distribution RMSE about 0.0044;
- baseline start-month predictive adequacy — **passed for calendar months, failed for serial dependence**; 11 of 12 month shares and the whole-year revision rate are reproduced, but exact continuations remain too low and other revisions too high; and
- full 29-parameter observed-information inference — **pending**; the gross-only January-heaping estimate is effectively on the boundary and should not receive a regular Wald interpretation;
- exact-anchor retention nesting and conditional identification — **passed**; zero reproduces the flexible baseline likelihood exactly, the empirical profile has a clear interior maximum, and 20

- conditional recovery samples recover the generating probability with small mean bias;
- exact-anchor predictive adequacy — **failed but improved**; exact continuations rise from 60.6% to 61.8% against 64.7% observed and other revisions fall from 32.4% to 31.4% against 28.3% observed;
 - longer 30-parameter multistart refinement — **passed provisionally**; low- and high-retention screens are decisively inferior, a second joint block adds only 1.71 likelihood points, and the final conditional polish adds 0.001 point, although the capped joint blocks retain convergence code 1;
 - full 30-parameter recovery — **passed for estimation and headline transition rates, inconclusive for the reporting decomposition**; three 5,000-record samples give entry- and exit-rate RMSEs of about 0.12 and 0.11 percentage points, while retention and the two revision probabilities remain noisy and offset one another;
 - full-rank OPG analytical inference — **passed provisionally**; the exact retention SE is 0.574 percentage points, but the information matrix is ill-conditioned, retention correlates 0.885 with tenure contamination, and gross-only January heaping is a boundary quantity; and
 - exact central observed-information inference — **pending**, with 48 of 1,801 likelihood points cached;
 - local-revision normalization and exact nesting — **passed**; the one-to-six-month kernel renormalizes at zero tenure, excludes zero from its support, and setting its probability to zero reproduces the exact-anchor log likelihood exactly;
 - local-revision conditional identification and multistart screening — **passed provisionally**; the profile has a clear interior maximum, low and high starts are inferior, and central reporting-block refinement improves the preferred basin, although the deliberately capped blocks retain convergence code 1; and
 - local-revision predictive adequacy — **failed but substantially improved**; the predicted one-to-six-month share rises from 3.0% to 10.8% against 11.4% observed, and the likelihood gain is concentrated in records with local changes, but exact and residual-revision moments remain outside the simulation intervals;
 - shared-reliability exact nesting and dispersed starts — **passed provisionally**; zero dispersion reproduces the independent likelihood, all starts improve on it, and the preferred basin is clearly superior, although the short joint blocks terminate at their iteration caps;
 - shared-reliability predictive adequacy — **failed but informative**; the local-revision and zero/one-irregularity predictions improve, but the model overpredicts two irregularities and leaves several duration moments outside their simulation intervals; and
 - cross-clock reliability interpretation — **limited by the three-wave design**; within-clock clustering identifies the dispersions, but the data do not directly reveal repeated tenure and repeated timegap clocks for the same person;
 - separate-dispersion joint convergence — **passed**; broad reporting and transition blocks materially improve the initial fit, two cached joint continuations add 78.9 and 49.4 likelihood points, and a third returns the identical parameter vector with convergence code zero;
 - fair common-dispersion refinement — **passed**; projecting the converged nuisance block into the restricted model, polishing the common shift, and checking the full restricted score reduces the equality-restriction LR to 144.6 without altering the transition rates;
 - separate-dispersion recovery — **passed provisionally**; one 10,000-person conditional recovery sample gives the same optimum from three starts and good latent-class discrimination, but more replications and full-nuisance recovery are still required;
 - separate-dispersion OPG inference — **passed numerically but limited for the restriction contrast**; the 33-parameter matrix is positive definite and full rank, but has condition

number about 55 million, one hazard direction needs a smaller perturbation, and its Wald approximation conflicts with the directly optimized LR test; and

- separate-dispersion nuisance-adjusted profile inference — **passed**; all retained points converge, two-start screens select the higher-likelihood warm branch on both sides, and the 95% profile interval for the timegap-minus-tenure dispersion is about 0.1100 to 0.1520, excluding the equality restriction and resolving the OPG/LR conflict in favor of the direct likelihood evidence; and
- separate-dispersion predictive adequacy — **failed**; exact continuations improve modestly, but local revisions, residual revisions, timegap inconsistencies, and zero/one/two-irregularity shares remain outside the simulation intervals.

8. Development log

Date	Change	Evidence or decision
2026-09-04	Added start-date reconstruction and A/B/C audit	Current-job start dates are persistent but frequently revised; January and whole-year patterns are prominent; strict job resets are visible.
2026-09-04	Selected the first model extension	Add a one-parameter employed-to-employed job-change mixture before enriching the reporting-error or AR(2) processes.
2026-09-04	Implemented and estimated the job-change mixture	The joint fit converged at ($q=0.0223$); fit improves sharply and estimated gross tenure contamination declines.
2026-09-04	Re-estimated on Panels B and C	The reset probability is stable in B and somewhat lower in C, while all three contamination estimates decline with stricter matching.
2026-09-04	Completed numerical validation	The preferred mode is stable and tightly profiled, but full-model recovery fails because the continuous likelihood mixes duration components with incompatible units.
2026-09-04	Replaced continuous densities with monthly masses	The discrete likelihood normalizes, substantially improves recovery, and estimates a Panel A reset probability near 0.97%.

Date	Change	Evidence or decision
2026-09-04	Repeated discrete-model recovery	Six fits at two sample sizes converge; bias and RMSE contract sharply at 40,000 observations and dispersed starts do not find a better mode.
2026-09-04	Completed discrete-model inference	The observed information is full rank and positive definite; analytical and nuisance-adjusted profile intervals agree closely.
2026-09-04	Re-estimated the corrected model on Panels B and C	All six multistart fits converge; reset and contamination estimates decline as matching becomes stricter, but the reset process remains present.
2026-09-04	Estimated persistent gross start-date reporting	The extension nests exactly and can recover positive persistence in simulation, but Panel A places the MLE at zero and leaves all substantive rates unchanged.
2026-09-04	Estimated January-heaped gross reporting	The normalized one-parameter extension passes recovery checks, all three Panel A starts agree near 24%, fit improves sharply, and the main transition estimates change little.
2026-09-04	Added analytical inference for January heaping	The 16-parameter observed-information matrix is full rank and positive definite; the main rates and heaping parameter are precise.
2026-09-04	Estimated correlated whole-year date revisions	The normalized extension passes recovery and multistart checks; its empirical implications are reported above.
2026-09-04	Completed revision-model inference and predictive diagnostics	Local information and profile inference are regular, but the fitted model does not reproduce the calendar moments and shifts likelihood away from non-target patterns.

Date	Change	Evidence or decision
2026-09-04	Allowed gross reports to revise preceding clean dates	Three starts converge near a 9.6% clean-anchor revision rate; fit and the targeted predictive moment improve sharply, but missing baseline start-month seasonality remains.
2026-09-04	Estimated a shared flexible start-month baseline	The likelihood improves by about 42,982 points, January receives 26.46% baseline mass, gross-only January heaping collapses to zero, and transition estimates remain stable.
2026-09-04	Validated the start-month extension	Distant starts meet and joint recovery succeeds after a one-sided gradient safeguard; calendar predictions pass, while remaining failures isolate serial report dependence.
2026-09-04	Added exact reported-start-date retention	The nested channel is strongly supported and conditionally recoverable; it improves exact-continuation predictions without materially changing entry or exit rates, but does not remove the full serial-reporting gap.
2026-09-04	Refined and jointly validated exact retention	Longer multistart work stabilizes the empirical point estimate near 20%; three full re-estimations recover pooled transitions reasonably well but reveal weak finite-sample separation among retention and revision probabilities.
2026-09-04	Added analytical inference diagnostics	The efficient OPG approximation is full rank and gives tight errors for headline transitions, while its condition number, reporting-channel correlations, and boundary heaping estimate document the remaining limitations.

Date	Change	Evidence or decision
2026-09-04	Estimated normalized local start-date revisions	The conditional probability is about 33.8%, multistart reporting blocks favor the same basin, and the predicted local-revision share moves from 3.0% to 10.8% against 11.4% observed without materially changing entry or exit rates.
2026-09-04	Added shared person-level duration-reporting reliability	The one-parameter equal-weight mixture strongly improves fit and duration-pattern clustering while leaving transition estimates stable, but its common cross-clock dispersion is a maintained restriction and predictive gaps remain.
2026-09-04	Relaxed equal tenure and timegap reliability dispersion	Symmetric restricted/unrestricted refinement converges at dispersions 2.09 and 2.22 with LR 144.6; transition rates remain stable, conditional recovery succeeds, and full-nuisance OPG errors are informative for headline rates but unreliable for the restriction contrast.
2026-09-04	Profiled the clock-specific dispersion difference	Multistart nuisance-adjusted profiling gives a 95% interval of about 0.1100 to 0.1520 for timegap minus tenure dispersion, decisively excluding equality and resolving the conflict with the ill-conditioned OPG Wald interval.

9. Four-wave AR(1) bridge to the AR(2) extension

The next substantive extension is employment-history dependence, rather than further refinement of the separation between duration-reliability classes. The four-wave bridge retains the preferred three-wave measurement model: discrete-month tenure, piecewise duration hazards, current-job resets, calendar and local reporting revisions, joint timegap contamination, and equal prior weights for the two reliability classes. No additional employment heterogeneity class has been introduced.

`EM-tenure/R/four_wave_eps.R` sums over all 16 four-wave latent employment histories. `EM-tenure/estimate_monthly_four_wave_ar1.R` prepares Panel A, caches incremental optimisation blocks, and constructs a waves 1–3 comparison using exactly the people and weights in the four-wave sample. The initial employment probability remains free. The likelihood is estimated directly, not with the old three-wave EM sufficient-statistic updates.

Sample and estimates

Subsequent validation warning: Section 10 identifies a timegap likelihood normalization defect, repaired in Section 11. The estimates below remain saved, numerically converged pre-repair results. Section 12 reports the separate empirical refit under the repaired likelihood when validated.

There are 317,437 records with four observed employment statuses and 316,889 with the required state-appropriate duration reports (548 exclusions). The likelihood collapses these to 101,916 distinct observation cells without changing their weighted contribution. Zero-month tenure is retained; unreported-state durations remain missing. Never-worked reports follow the same recoding convention as in the three-wave model. Interview months remain nominal quarter-end months, not recovered exact interview dates.

Table 46: Four-wave AR(1): numerical refinement and starting-point checks

Measure	Preliminary	Refined	Start (-)	Start (+)
Entry rate	0.02446	0.02603	0.02603	0.02603
Exit rate	0.04033	0.04060	0.04060	0.04060
Misclassification rate	0.02822	0.02765	0.02765	0.02765
Initial employment	0.48685	0.48705	0.48704	0.48705
Log likelihood	-2,890,792.87	-2,890,127.08	-2,890,127.11	-2,890,127.05

The preliminary column is an unconverged checkpoint. All three completed refinement runs have optimizer code zero and final maximum projected central scores between 1.5×10^{-6} and 3.0×10^{-6} per observation, below the declared 10^{-5} criterion. The optimizer’s scaled tolerance is chosen to imply this unscaled criterion; the latter is checked separately after termination. Start (+) has the highest log likelihood and is selected in `converged_comparison_latest.rds`. The three likelihoods differ by at most 0.056; entry, exit, and status-error rates differ by less than 0.00004 percentage points. This supports numerical stability of these rates, not precise identification of every reporting parameter.

Entry and exit rates pool the three quarterly intervals, weighting each individual-interval by the survey weight and posterior probability of the relevant origin state. Misclassification is the common

employment-status error probability, not a duration-contamination probability. The reliability mixture’s tenure and timegap midpoint parameters should not be interpreted as mean contamination probabilities.

Relative to the preliminary four-wave point, quarterly entry rises from 2.446% to 2.603%, exit rises slightly from 4.033% to 4.060%, status misclassification falls from 2.822% to 2.765%, and initial employment is essentially unchanged (48.685% versus 48.705%). The log likelihood improves by about 665.8. This is a numerical improvement within the **same model and sample**, not a test of a new specification. The preliminary fit understated entry more than exit; the broad finding of substantial misclassification remains. Likelihood levels on different three-wave and four-wave samples must not be compared as an LR test. The optional same-person three-wave re-estimation is omitted at the author’s request.

Checks and outstanding work

The continuation now has a separately validated compiled evaluator (`four_wave_fast.R` and `src/four_wave_monthly.cpp`). It uses the reference R probability tables and caches repeated spell and reporting-pattern calculations; no likelihood terms or model restrictions are changed. On the saved full sample, the log-likelihood difference is less than 10^{-7} and maximum posterior differences are below 10^{-13} . Perturbed-parameter, zero-status-error, disabled-revision, and serial/parallel comparisons also pass. A benchmark using eight threads takes about 1.3 seconds per full likelihood evaluation, rather than roughly a minute for the earlier R implementation.

`refine_four_wave_ar1_fast.R` jointly refines all 33 parameters, retains improving checkpoints, and checks a central-difference projected score after the optimizer stops. The diagonal of the saved three-wave information matrix is used only for numerical parameter scaling; it is not a substitute for four-wave standard errors and requires no three-wave re-estimation. The same-person three-wave re-estimation has been dropped from the work plan at the author’s request.

For the final continuation, category-averaged and missing-duration risks are calculated with exact piecewise-exponential integrals. The integral splits at the duration-hazard knots and their quarter-shifted counterparts. Fifty tests against independent numerical integration and limiting cases pass. At the saved preliminary point this changes the total log likelihood by about -0.000037 and the largest posterior probability by less than 2×10^{-8} ; its purpose is to remove amplified quadrature noise from the numerical score, not to change the model. The three-wave estimator is not modified. Central finite differences are used for the final joint refinement.

- Tests pass for all 16 histories, posterior normalization, the zero-status-error limit, three-wave joint-timegap equivalence, and four-wave timegap probability normalization. Preparation tests cover zero tenure, never-worked recoding, missing flags, and exclusion of invalid state-appropriate durations.
- Sharing and compressing repeated job-segment calculations reproduces the saved full-sample likelihood and posterior probabilities exactly. This is a computational change, not a model change.
- Accepted bounded optimisation steps are flagged as non-converged. Early checkpoints with a zero code are corrected when consolidated; step acceptance is not evidence that the likelihood has been maximized.
- All three four-wave runs pass optimizer and central-score convergence checks. The full-sample likelihood and posterior agree with the R reference at the fitted parameters. Starting-point

checks are saved separately in `multistart_summary_latest.csv`; all fit objects and diagnostics are retained.

- The two nearby starts perturb all 33 transformed parameters by $+0.05 \sin(j)$ and $-0.05 \sin(j)$, respectively, from the saved `start_multistart_exact.rds` point. Continuations resume saved checkpoints without applying the perturbation again.
- Nearby-start agreement is a numerical stability check, not proof of a global optimum or strong identification of the reliability-class separation.
- The first four-wave simulation gate fails (Section 10); Section 11 records the repair and renewed validation. Analytical standard errors and the duration-model AR(2) extension have not been estimated.

10. Four-wave simulation-recovery gate

Changes and scope

- Extended the existing sequential simulator to four waves, retaining its three-wave defaults and adding the exact piecewise-risk option.
- Generated three independent samples of 20,000 records at the selected Panel A four-wave parameters (seeds 260904–260906). These are simulations, not additional empirical panels or bootstrap replications.
- Checked the mean record-level score at the known truth before fitting. Its Monte Carlo standard error is the sample standard deviation of the score divided by the square root of the simulated sample size.
- Added exhaustive probability-mass and timegap-clock checks. No empirical fit or likelihood formula was changed, and no simulation recovery fit was run after these gates failed.

Results

The short-duration entry-hazard score is negative in all three samples (individual standardized scores -5.18, -5.26, and -5.50). The pooled mean is -0.01225 with Monte Carlo standard error 0.001331: a standardized discrepancy of -9.20. This fails the conservative screening threshold of five Monte Carlo standard errors. It establishes a generator/likelihood mismatch, not a precisely quantified bias in the empirical entry rate.

Table 47: Generator/likelihood score check: 60,000 simulated records

Parameter	Mean score	Monte Carlo SE	Standardized score
Entry log-hazard: 0–3 months	-0.012250	0.001331	-9.20
Entry log-hazard: 1–3 years	0.006410	0.001419	4.52
Status-error logit	0.001224	0.001121	1.09
Initial-employment logit	0.003123	0.001992	1.57

An independent finite-support check exposes a defect in the likelihood itself. The illustrative nested case uses constant tenure hazards of 20 per year, uniform start months, no job resets or serial tenure-revision channels, initial employment 0.5, tenure contamination 0.2, and timegap hazards either constant at 0.4 or equal to (4, 1, 0.5, 0.3, 0.2). It enumerates all four-wave observed status/duration combinations with tenure 0–24 months, including all seven timegap categories: 1,048,576 cells.

Table 48: Exhaustive normalization check in a nested example

Duration-dependent entry	Status error	Timegap error	Total probability
No	0.15	0.30	1.000000000
Yes	0.00	0.30	1.000000000
Yes	0.15	Approximately zero	1.000000000
Yes	0.15	0.30	0.999327181

When duration dependence, status errors, and timegap contamination are all present, total mass is **0.999327180679**, rather than one. Increasing the tenure cutoff from 12 to 24 months leaves the discrepancy essentially unchanged. A conservative omitted-tail bound at 24 months is below 1.7×10^{-10} ; it cannot explain the missing mass of about 0.000673. The R reference and compiled implementation give the same result at the 12-month cutoff. Constant-hazard and near-zero-error controls sum to one within numerical/tail tolerance. This is a counterexample to normalization, not an estimate of the defect’s size at the empirical parameters.

Diagnosis and interpretation

- **Future-report-dependent fallback:** the four-wave E-step uses a joint timegap likelihood only when an entire latent nonemployment spell is observed as nonemployment. Otherwise it falls back to adjacent-pair terms. Thus a fourth reported employment status can change the likelihood factor for the first three timegap reports. In the illustrative case, the joint three-report distribution and pairwise fallback differ by 0.2286 in total variation. Individually normalized components do not guarantee a normalized full likelihood when transition risks also depend on the reports.
- **A separate simulator mismatch:** the simulator retains a continuous nonemployment clock and ages it across waves. The likelihood instead multiplies category-to-category transition probabilities, effectively discarding within-category information from earlier reports. Exact interval-intersection probabilities and that category-Markov approximation differ even with negligible timegap contamination (total variation 0.02143 in the same illustrative hazard case).
- These findings are about probability construction, not merely weak identification of the two reliability classes. They do not establish how much the empirical entry, exit, or misclassification rates will change. Earlier numerical convergence and implementation-equivalence checks still hold, but the four-wave results are not yet validated for interpretation. Earlier normalization checks covered individual emissions and posterior weights, not the sum of the complete joint probability over observable records; those are different tests.

The repair should use one joint timegap-spell likelihood for every observed subset, integrating missing or contaminated reports and retaining the clean clock’s interval restrictions across all waves. The simulator must implement that same probability law. Re-run normalization and score checks first; only then run the small parameter-recovery fits. This repair was deferred at the initial validation stage and subsequently authorized; see Section 11.

Reproduction scripts are `validate_four_wave_simulation.R`, `audit_four_wave_probability_mass.R`, and `summarise_four_wave_recovery.R`. For the mass audit, use `FOUR_WAVE_MASS_MAX_MONTH=24` for the tight-tail check; use 12 with `FOUR_WAVE_MASS_REFERENCE=true` for the independent R

comparison. Set `FOUR_WAVE_TIMEGAP_CLOCK=legacy_pairwise` to reproduce these historical failures. The consolidated `recovery/recovery_status_latest.rds` records the failed gates, zero recovery fits, and unchanged empirical estimates.

11. Repaired continuous timegap clock and recovery

Changes

- Replaced the four-wave joint/pairwise fallback with one joint timegap likelihood for every latent nonemployment spell and every observed subset.
- Each clean report restricts the same spell-start duration. Missing reports are integrated out, and all contamination patterns are summed. Information from earlier categories is retained rather than discarded at each wave.
- Applied the same calculation in the R reference and compiled evaluator. The existing sequential simulator already retains and ages this clock; its four-wave exact-risk option now matches the repaired likelihood.
- Kept the initial timegap distribution, transition-risk formulas, tenure reporting model, and three-wave estimator unchanged. In particular, this repair does not introduce a new within-quarter distribution for timegap at a nonemployment spell's onset. That remains a separate modelling question.
- Preserved legacy evaluation explicitly as `legacy_pairwise`; repaired four-wave evaluation defaults to `continuous_joint`. Historical fit objects and failed diagnostics are not overwritten.

For a clean report in category interval $[a_t, b_t)$ at offset t , the spell-start clock satisfies $D_0 \in [a_t - t\Delta, b_t - t\Delta)$. Its joint clean-report probability is

$$P \left\{ \max(0, \max_t(a_t - t\Delta)) \leq D_0 < \min_t(b_t - t\Delta) \right\}, \quad \Delta = 0.25.$$

An empty intersection has probability zero. Gross reports instead contribute their marginal category probabilities; unobserved reports impose no restriction. Mixture weights retain the existing clock-specific reliability classes. This enforces clean-clock consistency across all observed waves without requiring every status report to be correct.

Probability and score checks

All eight exhaustive probability cases now sum to one within numerical tolerance, including the former failing case. Summing out any report gives the same joint distribution for the remaining reports. Fifty-six new unit checks cover observed subsets, boundary contamination probabilities, interval intersections, and missing early reports. The repaired full-sample compiled likelihood agrees with R within 4.6×10^{-8} ; maximum posterior differences are below 1.3×10^{-14} .

Repeating the same three 20,000-person simulation samples removes the score failure: the short-duration entry-hazard discrepancy falls from -9.20 to +1.86 Monte Carlo standard errors. No pooled score among the 33 parameters exceeds 1.87 in absolute value. These checks support the repaired probability law; they do not by themselves establish finite-sample recovery or precision.

Table 49: Standardized score at the generating parameters:
before and after repair

Parameter	Legacy	Repaired
Entry log-hazard: 0–3 months	-9.20	1.86
Entry log-hazard: 1–3 years	4.52	1.36
Status-error logit	1.09	1.46
Initial-employment logit	1.57	1.46

Small full-parameter recovery exercise

`estimate_four_wave_recovery.R` estimates all 33 parameters on two independent 5,000-person samples, using two nearby starting points for the first sample and one for the second (seeds 261004 and 261005). It uses the same bounded likelihood optimization and a separate projected central-score tolerance of 10^{-5} . The saved empirical scaling vector is only a numerical preconditioner. No nuisance parameter is fixed at its generating value.

Table 50: Simulation recovery: sample 1

Measure	Generating	near	alternative
Entry rate	0.02936	0.02785	0.02785
Exit rate	0.04155	0.04135	0.04135
Misclassification rate	0.02765	0.02855	0.02855
Initial employment	0.48705	0.47605	0.47605

Table 51: Simulation recovery: sample 2

Measure	Generating	near
Entry rate	0.02936	0.02909
Exit rate	0.04139	0.04189
Misclassification rate	0.02765	0.02860
Initial employment	0.48705	0.48865

Table 52: Recovery convergence diagnostics

replication	start	n	converged	score	loglik
1	near	5000	TRUE	2.54e-06	-46198.28
1	alternative	5000	TRUE	2.93e-06	-46198.28
2	near	5000	TRUE	2.84e-06	-46827.44

The generating transition rates use the true parameters and their posterior risk weights on the same simulated sample, so differences isolate parameter estimation rather than differences in sample composition. Full hazard coefficients are saved in `coefficient_recovery_latest.csv` and shown

below; fit objects also retain all reporting parameters and numerical diagnostics. This is a small implementation/recovery check, not a large Monte Carlo bias study or a bootstrap standard-error exercise. Empirical Panel A rates are not updated by these simulated fits.

All three fits converge with maximum projected central scores between 2.54×10^{-6} and 2.93×10^{-6} . The two starts for sample 1 differ by less than 0.001 log-likelihood units and 0.00013 percentage points in the main rates. Across the two samples, entry, exit, and status misclassification are each within 0.16 percentage points of their generating values. Initial employment differs by -1.10 and +0.16 percentage points, respectively. The high short-duration entry hazard and lower long-duration hazards remain visible in the recovered coefficients. These results support the repaired implementation and approximate recovery of economically relevant quantities; two samples do not establish finite-sample unbiasedness, precise reliability separation, or a globally unique optimum.

Table 53: Duration-hazard recovery (annual hazards, not quarterly transition probabilities)

Hazard	Generating	Sample 1	Sample 2
Entry: 0–3 months	0.5088	0.5085	0.4835
Entry: 3–12 months	0.2399	0.2391	0.2492
Entry: 1–3 years	0.0941	0.0893	0.0941
Entry: 3–5 years	0.1010	0.0967	0.1003
Entry: 5+ years	0.0719	0.0630	0.0683
Exit: 0–3 months	0.2532	0.2809	0.2353
Exit: 3–12 months	0.2869	0.2880	0.2784
Exit: 1–3 years	0.1627	0.1597	0.1657
Exit: 3–5 years	0.1448	0.1431	0.1541
Exit: 5+ years	0.1286	0.1265	0.1331

New outputs are under `four_wave_ar1/recovery_continuous_clock`; the failed legacy checks remain under `four_wave_ar1/recovery`. The empirical continuation defaults to a new `clock_joint1` label and rejects stale-source checkpoints.

12. Repaired four-wave AR(1): empirical Panel A refit

Changes and scope

- Re-estimate all 33 parameters under the repaired continuous-clock likelihood on the same 316,889 Panel A records (101,916 collapsed likelihood cells).
- Start at the best converged pre-repair fit, preserve the old fit, and use a separate output label. Require optimizer convergence and an independent projected central score no greater than 10^{-5} per unit of sample weight.
- Keep the duration hazards, tenure reporting specification, fixed equal reliability-class shares, and free initial employment probability unchanged. Do not re-estimate the three-wave comparator or introduce AR(2) in this step.
- Compare quarterly entry and exit probabilities using the same pooled posterior-origin-risk and survey weights. Misclassification refers to employment-status error, not duration contamination.

Table 54: Panel A four-wave AR(1): main rates (percent)

Measure	Pre-repair	Repaired	Change (pp)
Entry rate	2.603	2.693	0.090
Exit rate	4.060	4.092	0.031
Misclassification rate	2.765	2.646	-0.119
Initial employment	48.705	49.108	0.403

Repaired log likelihood: -2894475.050; maximum projected central score: 5.53e-06.

Re-optimization improves the repaired objective by 414.246 relative to the old parameter vector.

Table 55: Panel A duration coefficients: annual hazards

Hazard	Duration	Pre-repair	Repaired
Entry	0–3 months	0.5088	0.5635
Entry	3–12 months	0.2399	0.2404
Entry	1–3 years	0.0941	0.0916
Entry	3–5 years	0.1010	0.1017
Entry	5+ years	0.0719	0.0767
Exit	0–3 months	0.2532	0.2521
Exit	3–12 months	0.2869	0.2910
Exit	1–3 years	0.1627	0.1635
Exit	3–5 years	0.1448	0.1453
Exit	5+ years	0.1286	0.1289

Findings and interpretation

- The repaired empirical fit converges with optimizer code zero and a separately recomputed maximum projected central score of 5.53×10^{-6} . All 33 parameters were optimized from the best converged legacy starting vector. This is one empirical starting point, not a new dispersed-start exercise.
- Entry rises from 2.603% to 2.693% per quarter; exit rises from 4.060% to 4.092%. Employment-status misclassification falls from 2.765% to 2.646%, and initial employment rises from 48.705% to 49.108%.
- Thus the repair changes the estimates modestly rather than overturning their main interpretation. Estimated status misclassification remains material, while exit probability conditional on employment remains above entry probability conditional on nonemployment. Those probabilities have different populations at risk and should not be directly subtracted to infer the change in employment.
- The largest duration-hazard adjustment is in the shortest nonemployment band: its annual entry hazard rises from about 0.509 to 0.563. Longer-duration entry hazards remain substantially lower; exit hazards change comparatively little. These are annual hazard coefficients, not the weighted quarterly entry probabilities in the main comparison.

The old likelihood is retained for historical auditing, not as a nested-model benchmark: changing a defective probability construction means that its difference from the repaired likelihood is not an

LR statistic. The gain reported above compares two parameter vectors under the *same repaired* objective. Neither convergence nor the recovery checks establish strong identification of the two reliability dispersions or a globally unique optimum. No analytical or bootstrapped standard errors are estimated in this step.

At the new fitted parameter vector, the independent full-sample R evaluator agrees with the compiled likelihood within 4.52×10^{-8} and with its posteriors within 1.24×10^{-14} . The repaired log likelihood is -2,894,475.050, a gain of 414.246 over evaluating the old parameter vector under the repaired law. The focused regression suite passes 249 checks.

Execution note: a trailing parsing error was reported after the optimizer, independent score check, fit save, and rate printing had completed; the runner had been edited while active. The current runner parses cleanly. A fresh process loaded the saved fit and successfully performed the independent full-sample checks and consolidation above. The error did not require a new estimation or affect the saved parameter estimates.

Run `refine_four_wave_ar1_fast.R` with label `clock_joint1`, using `fit_fast_start_plus_latest.rds` as the start, then run `summarise_four_wave_clock_empirical.R`. The collector verifies the full-sample R reference at the new parameters and stores rates, annual hazards, all transformed coefficients, posteriors, convergence diagnostics, and source/input fingerprints in `four_wave_ar1/continuous_clock_empirical`. These files do not replace the legacy results or the separate simulation-recovery outputs.

13. Four-wave duration model with two AR(2) return effects

A compact candidate paper Table 9, including history-specific entry and exit probabilities, is provided in `four_wave_duration_results.Rmd`. Rebuild its LaTeX and unrounded CSV snapshot with `scripts/build_four_wave_duration_paper_table.R`; this reads the saved fits without re-estimation. Existing paper Tables 1–8 are unchanged. Analytical uncertainty and the questionnaire-motivated sensitivity checks remain pending.

Changes and identifying restrictions

- Add one entry and one exit coefficient to the repaired Panel A model; re-estimate all 35 parameters jointly. Keep duration-reporting mechanisms, duration reliability classes, and the estimation sample unchanged.
- For transitions 2-to-3 and 3-to-4, shift the log odds of a transition when the current latent state differs from the preceding latent state. Positive coefficients mean an increased probability of returning to the prior state.
- Retain free initial employment and the AR(1) law for transition 1-to-2. This is an explicit initial-pair restriction, not an assertion of stationarity or an estimate of unobserved pre-panel history. A free initial-pair model is deferred to a sensitivity check.
- Use forward numerical gradients during optimization and an independent central score for acceptance. Every accepted fit must have optimizer code zero and maximum projected central score at most 10^{-5} . Use a main start and two perturbed starts, with all nuisance parameters free.

Writing $S_t = 1$ for employment and $r_{01,t}, r_{10,t}$ for the existing duration-dependent AR(1) probabilities, for $t = 2, 3$:

$$\begin{aligned}\text{logit } P(S_{t+1} = 1 \mid S_t = 0, \dots) &= \text{logit}(r_{01,t}) + b_N 1\{S_{t-1} = 1\}, \\ \text{logit } P(S_{t+1} = 0 \mid S_t = 1, \dots) &= \text{logit}(r_{10,t}) + b_E 1\{S_{t-1} = 0\}.\end{aligned}$$

Thus $\exp(b_N)$ is the entry-odds multiplier after a recent exit and $\exp(b_E)$ is the exit-odds multiplier after a recent entry, holding the baseline duration-dependent probability fixed. These are **odds ratios, not hazard ratios**. Apply them after the existing integration over category or unobserved wrong-state durations. Do not alter the marginal duration-reporting distributions or impose a new stationary duration distribution.

The implementation reuses the validated AR(1) likelihood through the exact identity $L_2 = L_1 \sum_h P_1(h \mid \text{reports}) P_2(h) / P_1(h)$. The AR(1) likelihood and posterior are recomputed at every new nuisance vector: neither is fixed during estimation. The independent R reference instead constructs the AR(2) history prior directly before combining reporting probabilities. Class and job-change posterior aggregates must also be recomputed under AR(2); they cannot simply be copied from the AR(1) evaluator.

Validation and recovery

Zero effects reproduce the full-sample repaired AR(1) likelihood and posteriors. Nonzero effects agree with the independent direct-prior R calculation. Four exhaustive finite-support joint-probability checks sum to one, within the tenure-tail allowance. Three 10,000-person simulations check all 35 likelihood scores at known nonzero coefficients. The largest pooled absolute discrepancy is 2.18 Monte Carlo standard errors; the entry and exit AR(2) discrepancies are 0.83 and -0.01. Twenty-four added unit checks cover nesting, reference agreement, the seeded simulator, and which histories receive each effect. These checks concern implementation, not precision of empirical effects.

Table 56: AR(2) recovery with all 35 parameters free: two 5,000-person samples

replication	start	converged	ar2_entry_truth	ar2_entry	ar2_exit_truth	ar2_exit
1	near	TRUE	1.0986	1.2447	0.6931	0.4720
1	alternative	TRUE	1.0986	1.2448	0.6931	0.4722
2	near	TRUE	1.0986	0.9640	0.6931	0.6197

Empirical comparison

All three full-parameter recovery fits converge, with independent projected central scores below 4.4×10^{-6} . The two starts on sample 1 differ by less than 0.0004 likelihood units and 0.00024 in either AR(2) coefficient. The generating entry coefficient is 1.099; estimates are 1.245 and 0.964 across the two samples. The generating exit coefficient is 0.693; estimates are 0.472 and 0.620. These are encouraging implementation checks, not a large-sample bias or coverage study. After projecting out all nuisance-score directions in the 30,000-person score simulation, the two-by-two information matrix has eigenvalues 70.78 and 56.65: the two return effects have separately varying scores in this experiment.

Table 57: Repaired four-wave duration model: AR(1) and AR(2), Panel A

Measure	AR1	AR2
Entry rate (%)	2.693	3.206
Exit rate (%)	4.092	4.775
Misclassification rate (%)	2.646	2.441
Initial employment (%)	49.108	49.102
Log likelihood	-2,894,475.050	-2,892,278.647
N	316,889	316,889

Table 58: Return effects: both coefficients are fixed at zero in AR(1)

Measure	AR1	AR2
Entry return effect (log odds)	0	1.299
Exit return effect (log odds)	0	1.931
Entry return odds ratio	1	3.666
Exit return odds ratio	1	6.895

Log-likelihood gain: 2196.404; twice the gain: 4392.807.

Table 59: AR(2) starting-point and convergence checks

model	loglik	score	ar2_entry	ar2_exit
primary	-2892279	7.70e-06	1.2987	1.9308
minus	-2892279	3.68e-06	1.2990	1.9308
plus	-2892279	3.72e-06	1.2990	1.9308

All main rates pool the three transitions using survey weights and posterior origin-risk weights; only the last two transition probabilities contain an AR(2) term. Initial employment is the estimated initial probability. The misclassification parameter concerns employment status, not duration reports. Unlike the historical pre-repair comparison, these two models are nested under one repaired probability law. Any formal LR inference still requires regularity and identification assumptions; the likelihood gain alone is not a validation of causal state dependence.

Table 60: Posterior decomposition of apparent entry transitions (%)

Measure	AR1	AR2
Raw reversal	35.68	35.68
Not confirmed	61.79	55.80
Genuine, reversed	6.48	13.99
Genuine, persistent	31.73	30.21

Table 61: Posterior decomposition of apparent exit transitions (%)

Measure	AR1	AR2
Raw reversal	34.55	34.55
Not confirmed	51.16	45.88
Genuine, reversed	5.54	10.06
Genuine, persistent	43.30	44.05

This decomposition pools transitions 1-to-2 and 2-to-3, which have an observed follow-up wave. A transition is confirmed only when the latent origin and destination match its apparent direction. The three model-based categories partition apparent transitions into not confirmed, genuine but reversed next wave, and genuine and persistent next wave. All four reports inform the posterior; this is a retrospective decomposition, not a forecast using just two reports. “Not confirmed” does not distinguish misreporting from record-matching errors.

Table 62: Baseline annual duration hazards: the AR(2) odds adjustment is additional

hazard	duration	AR1	AR2
Entry	0–3 months	0.5635	0.5577
Entry	3–12 months	0.2404	0.2406
Entry	1–3 years	0.0916	0.0922
Entry	3–5 years	0.1017	0.1022
Entry	5+ years	0.0767	0.0857
Exit	0–3 months	0.2521	0.2346
Exit	3–12 months	0.2910	0.2894
Exit	1–3 years	0.1635	0.1640
Exit	3–5 years	0.1453	0.1456
Exit	5+ years	0.1289	0.1290

Findings and interpretation

- All three empirical starts converge with independent projected central scores between 3.68×10^{-6} and 7.70×10^{-6} . Their entry, exit, and status-misclassification rates differ by less than 0.00007 percentage points, and their return coefficients by less than 0.00027. The highest-likelihood run is **plus**; its log likelihood exceeds repaired AR(1) by 2,196.404 (twice the gain is 4,392.807). The small likelihood spread across starts is 0.112, not evidence that every nuisance parameter is strongly identified or that a global optimum has been proved.
- The entry return coefficient is 1.299 and the exit return coefficient 1.931: odds multipliers of 3.666 and 6.895, respectively. These duration controls do not absorb the tendency to return to the preceding state. Current-job tenure is not the same as uninterrupted employment duration when workers can change jobs while remaining employed; this is one reason surviving history effects need not contradict duration dependence.
- Average quarterly entry rises from 2.693% to 3.206%, and exit from 4.092% to 4.775%. Status misclassification falls modestly, from 2.646% to 2.441%; initial employment barely changes

(49.108% versus 49.102%). The main interpretation is more genuine turnover, not a large revision to initial employment or the disappearance of misclassification.

- Among apparent entries, the share inferred to be genuine and reversed next wave rises from 6.48% to 13.99%; among apparent exits it rises from 5.54% to 10.06%. Persistent-transition shares change much less. Nevertheless, 55.80% of apparent entries and 45.88% of apparent exits are not confirmed in their reported direction under AR(2). The model therefore attributes more short-lived movement to genuine transitions while retaining an important role for misclassification, from either reporting or matching errors. These are shares of all apparent transitions, not a decomposition conditional on observing a reversal.
- After projecting out all 33 nuisance-score directions, the empirical two-effect information matrix has eigenvalues 1,226.75 and 1,634.83 (condition number 1.33), unchanged across three projection tolerances. This supports local separation of the two return effects; it does not remove the earlier caveat about weak duration-reliability separation. No parameter reaches its numerical bound in the selected fit.

The full-sample direct-prior R and reweighted compiled likelihoods agree to displayed machine precision; the maximum posterior difference is 3.84×10^{-15} . Analytical return-coefficient scores agree with the numerical scores within 4.3×10^{-11} . The weighted posterior mean status misclassification is 2.44142%, consistent with the common fitted probability. The focused suite passes 273 checks, including the audit table chunks.

Remaining checks

- Check sensitivity to the initial-pair restriction before a structural interpretation. Keep broader reliability-mixture refinements secondary.
- Retain the separate caveat from Section 11: the timegap clock at a newly entered nonemployment spell is still drawn from the existing marginal law, rather than restricted to an onset within the last quarter. The AR(2) extension does not repair or validate that substantive modelling assumption.
- Interpret any surviving AR(2) effects as conditional associations, potentially reflecting residual duration misspecification, omitted heterogeneity, or reporting assumptions. No analytical or bootstrap standard errors are computed in this first extension.

Reproduce with `validate_four_wave_ar2.R`, `recover_four_wave_ar2.R`, `estimate_four_wave_ar2.R`, and `summarise_four_wave_ar2.R`, in that order. Outputs reside in the four-wave AR(2) results directory; repaired AR(1) and historical pre-repair objects are preserved.

14. Subsequent extensions

The flexible baseline repairs the calendar distribution, exact retention and local revisions capture part of the serial dependence, and separate clock-specific reliability dispersions capture additional clustering of duration irregularities. Following the decision to limit further work on reliability-class separation, the remaining work should proceed in this order:

The repaired Panel A AR(1) refit is reported in Section 12. Its AR(2) extension, recovery, nearby-start checks, local-identification diagnostic, and first reversal comparison are completed in Section 13. The same-person three-wave re-estimation remains omitted at the author's request.

1. check sensitivity to the initial-pair rule and the clean timegap distribution at a new nonemployment spell, changing one assumption at a time;

2. if a structural interpretation of the return effects is required, assess finer short-duration controls and remaining observed or unobserved heterogeneity rather than treating AR(2) coefficients as causal effects;
3. compute approximate analytical uncertainty once the preferred specification is settled, treating weak nuisance directions explicitly; and
4. retain uncertainty about reliability-class separation as a limitation. Additional class-share and bootstrap sensitivity work is secondary rather than a prerequisite for the employment extension.